Learning Transferable Basin and River-Network Representations for Hydrologic Prediction Across Space and Time

How we represent a basin decides whether a model generalizes

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- Academic Background
- Research Overview
- Literature Review

Completed	**Study 1:** Static-Dynamic Fusion **Study 2:** BASINGRID Data **Study 3:** Basin2Vec Embeddings
Ongoing	**Study 4:** Graph River Network Future Roadmap at ORNL

- Conclusion

Academic Background

University of Engineering and Technology, Lahore (2017-2021)

- B.S. Computer Science

Michigan State University (2023-Present)

- Dual Ph. D.
  - Computer Science (advised by Dr. Pang-Ning Tan)
  - Biosystems and Agriculture Engineering (advised by Dr. A. Pouyan Nejadhashemi)

Research Overview

Motivation: Streamflow, what it is and why it matters

Streamflow is the **volume of water moving through a river** at a particular location over time.

- **Public safety**
  Flood warning, drought response, emergency planning
- **Agriculture & water supply**
  Irrigation demand, reservoir operation, water allocation
- **Ecosystems & infrastructure**
  Environmental flows, design standards, climate adaptation

NOAA reports losses of approximately $367.5 billion from drought and $203.0 billion from inland flooding over 1980–2024.

*Reliable streamflow information supports decisions under both **water excess** and **scarcity**.*

Motivation: The gauge gap

2.7 Million+

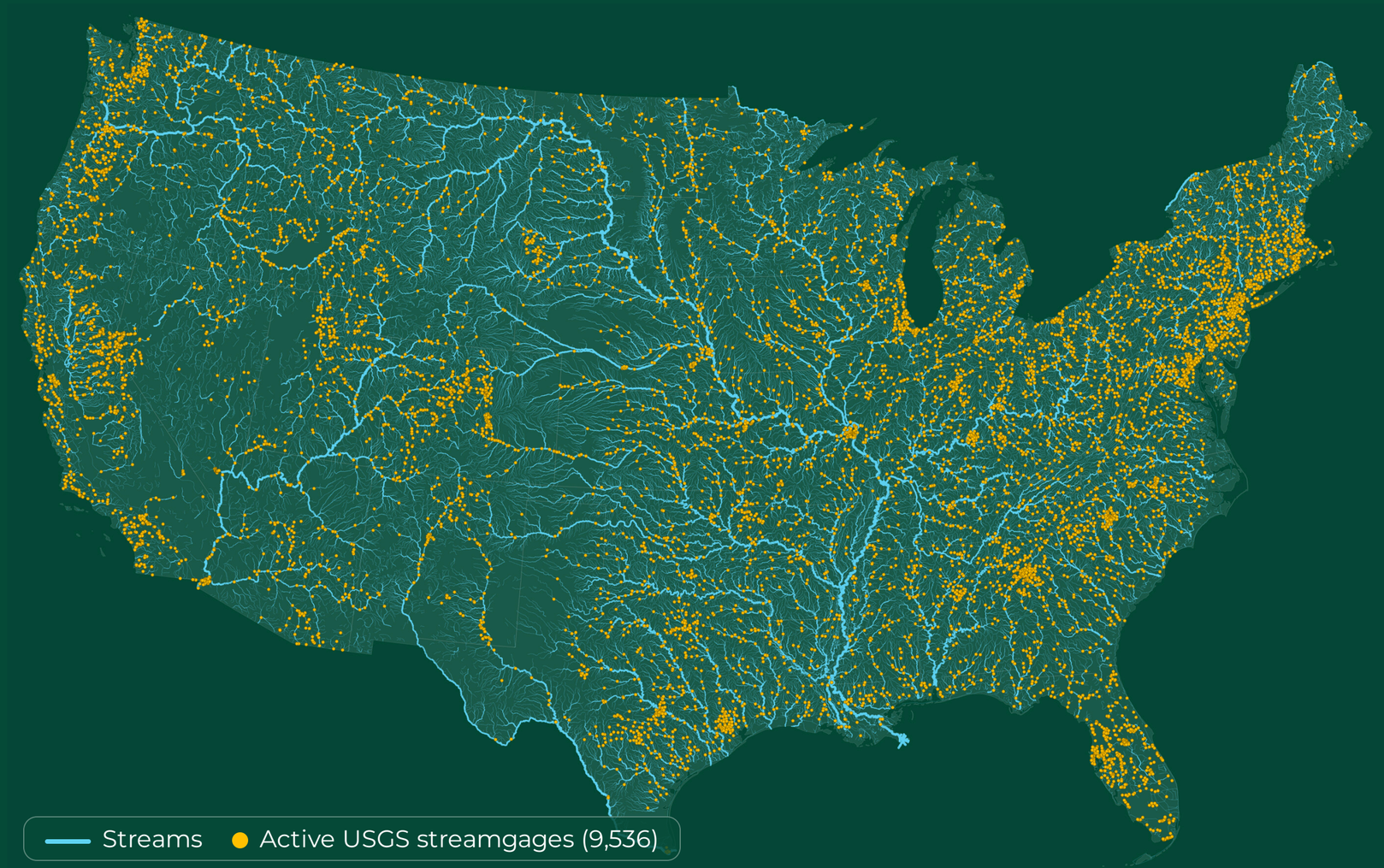
River reaches across the CONUS

~12k

Federal priority streamgage locations

~5k

Inactive gauges due to lack of funds



Gauges cannot observe every reach. Models must predict where we do not measure.

Rivers: USGS National Atlas 1:1M hydrography (lower 48). Gauges: USGS NWIS active stream sites reporting daily discharge, retrieved Nov 2026. Albers equal-area projection.

Monitoring is essential!

but gauge measurements **cannot directly observe** every part of the national river network.

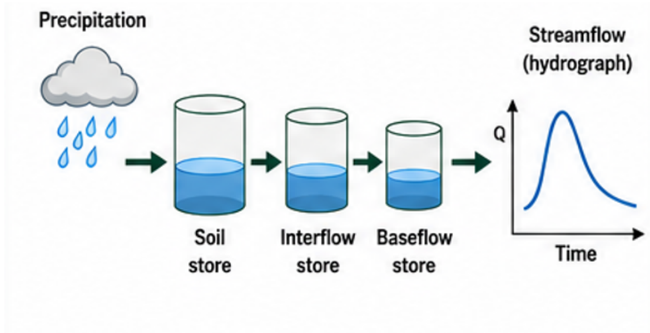
That's why **streamflow modeling** is needed

How is streamflow modeled?

Three broad model families represent the rainfall–runoff relationship in different ways.

1. Conceptual models

Simplified representation of watershed stores and fluxes.



Pros

- Hydrologically interpretable
- Simple structure, efficient to run
- Works with relatively limited data

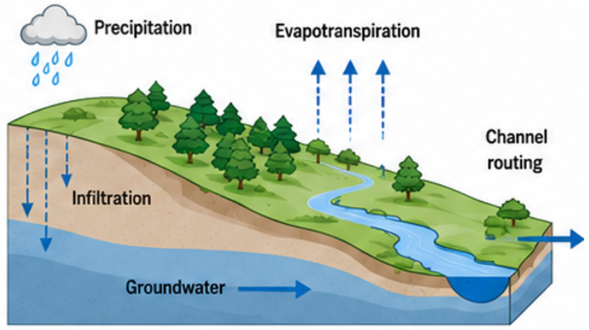
Cons

- Processes are simplified
- Requires calibration for each basin
- Limited ability to represent spatial variability

Examples: HBV, SAC-SMA, Xinanjiang, GR4J

2. Process-based models

Explicit representation of physical processes across the watershed.



Pros

- Physically based and interpretable
- Captures spatial and temporal processes in detail
- Can simulate many variables beyond streamflow

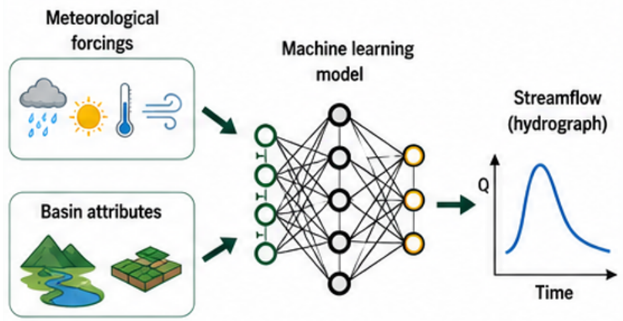
Cons

- Data intensive (inputs & parameters)
- Computationally expensive
- Difficult to calibrate and transfer across many basins

Examples: SWAT, VIC, MIKE SHE, HYPE, TOPKAPI

3. Data-driven models

Learn the forcing–streamflow relationship from observed data.



Pros

- Strong predictive performance
- Learns complex, nonlinear relationships
- Shares information across many basins (regional learning)

Cons

- Less interpretable
- Performance depends on quality of data and representation
- May extrapolate poorly outside training regimes

Examples: LSTM, EA-LSTM, GRU, Transformers, GNNs

Takeaway: Model family describes how the rainfall–runoff relationship is represented.
It is independent of the spatial setting (lumped vs. distributed) and the generalization task (temporal vs. spatial).

Modeling streamflow requires data.

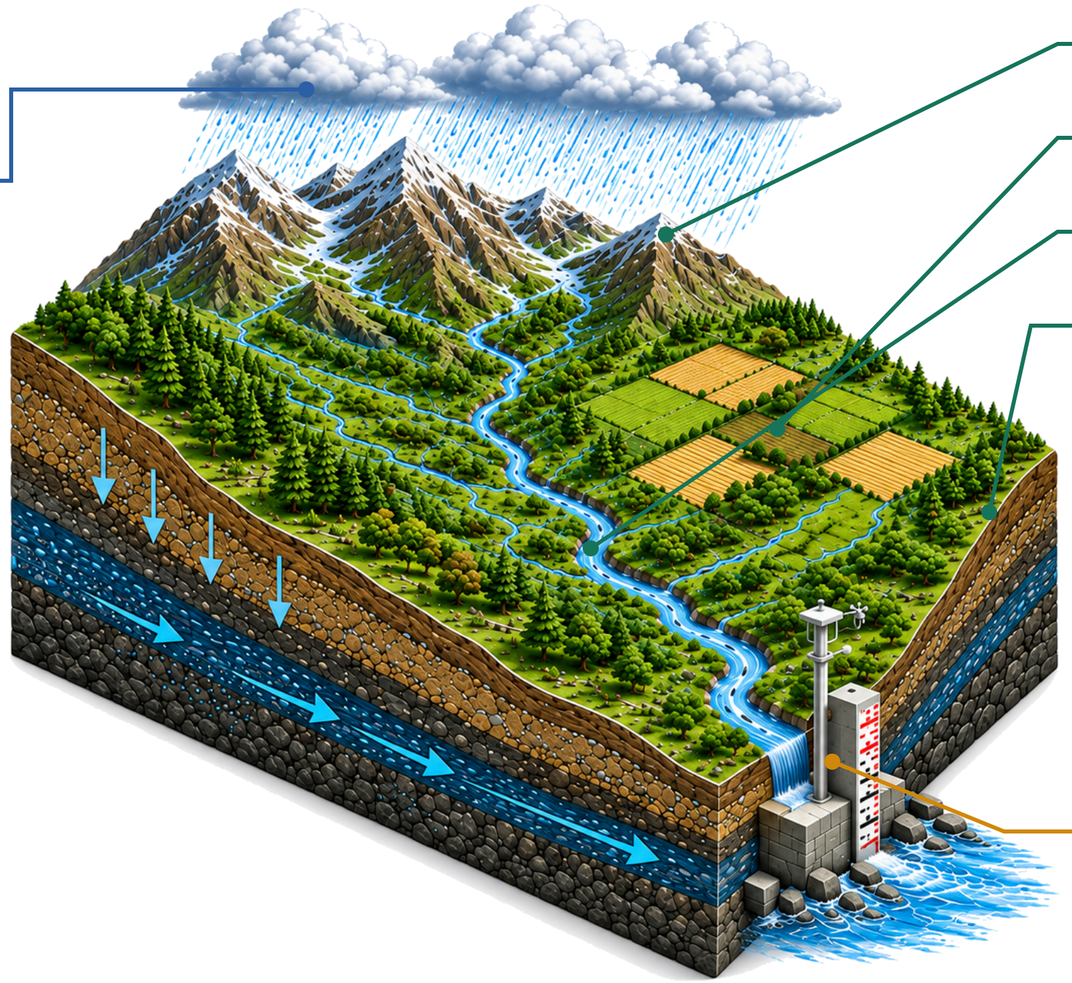
- Data types: **dynamic** and **static**

Data Types

DYNAMIC VARIABLES
(vary over time)

- **Precipitation**
  rainfall, snow
- **Temperature**
  min / max air temperature
- **Wind speed**
- **Humidity**
  vapor pressure
- **Air pressure**
- **Solar radiation**
  shortwave, day length

...

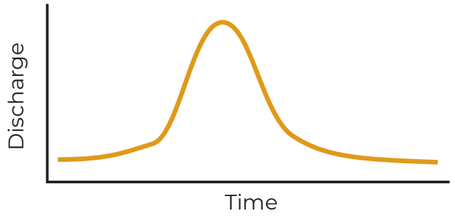


STATIC VARIABLES
(do not vary over time)

- **Terrain**
  elevation, slope
- **Land cover**
  vegetation, crops
- **Drainage network**
  river topology
- **Soils**
  soil type, texture
- **Geology**
  rock type
- **Hydrologic properties**
  porosity, Ksat, AWC

...

OBSERVED OUTPUT
Outlet discharge



Dynamic variables **drive** the hydrologic cycle
inputs that change over time

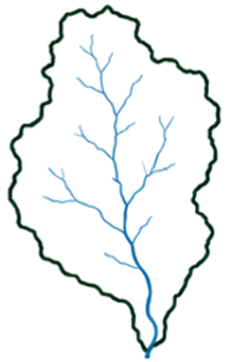
Static variables **define** catchment characteristics
inputs that remain constant

Modeling streamflow requires data.

- Data types: **dynamic** and **static**
- Data representation: **lumped** and **distributed**

Motivation: Lumped vs. distributed basin representations

Lumped Representation

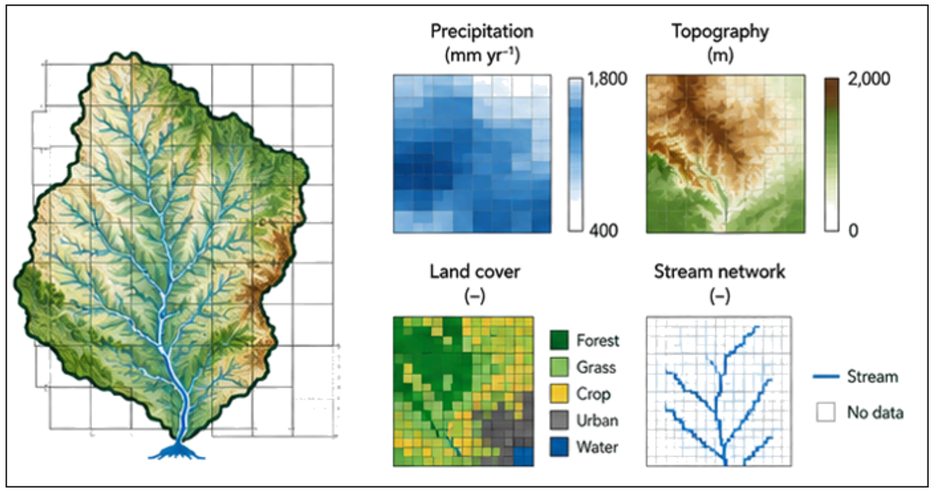


Aggregated attributes (feature vector)

Feature	Example
Area (km^2)	1,250
Mean precipitation ($mm\ yr^{-1}$)	1,120
Mean slope (%)	12.8
Forest fraction (–)	0.62
Soil texture (sand %)	42

Trade-off

Distributed Representation



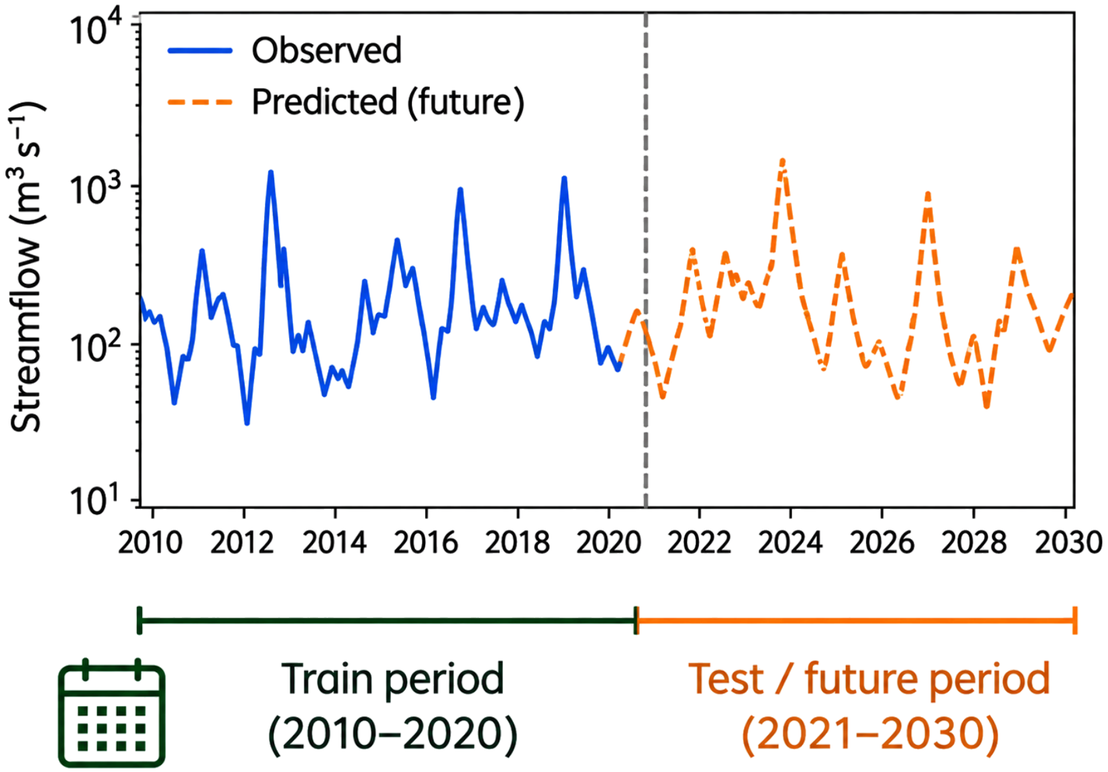
Benefits	Limitations
• Compact • Scalable • Easy to use	• **Spatial heterogeneity lost** • Local process information removed • Drainage connectivity simplified

Benefits	Limitations
• **Spatially rich** • Retains basin geometry • Keeps drainage structure	• Hard to align across sources • Expensive to process and store • **Hard to scale and standardize**

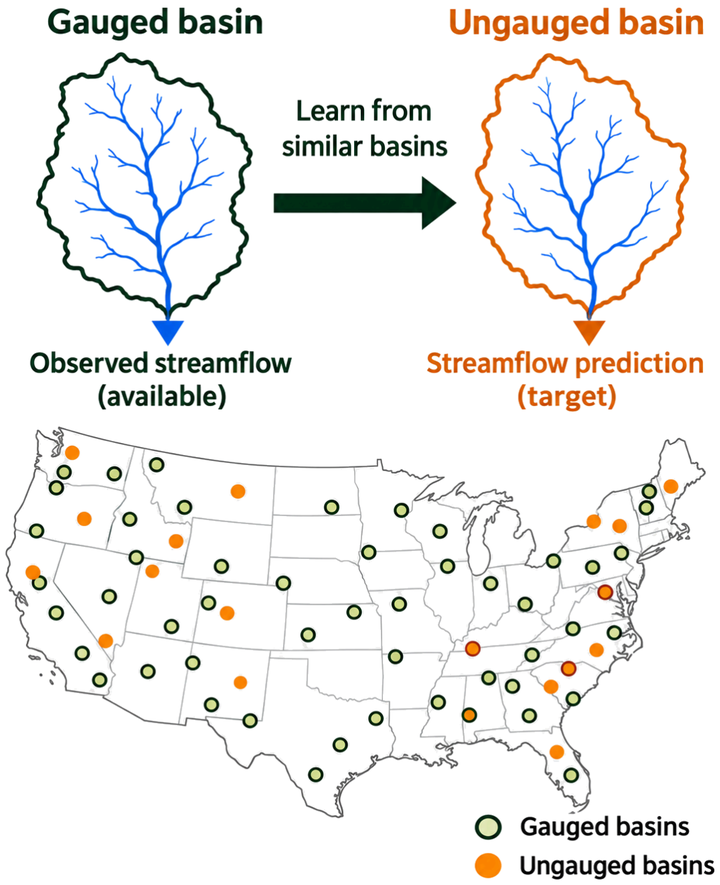
Representation must balance **compact** with **distributed spatial information**.

Motivation: Two tests of generalization, time and space

Temporal: same basins, future period



Spatial: basins or gauges never seen



Thesis: how we represent a basin decides whether a model generalizes.

Literature Review: What has been done, what limits it, what is missing

	Hydrologic Modeling at Scale	Large-Sample Datasets	Transferability
What has been done	Mostly **lumped** basin representations **LSTM, EA-LSTM** *Kratzert et al. 2018, 2019; Addor et al. 2017*	Standardized basin-averaged forcing and catchment attributes **CAMELS, Caravan** *Addor et al. 2017; Linke et al. 2019; Kratzert et al. 2023*	Earth-observation models learn compact representations from imagery and geospatial data **AlphaEarth, SatCLIP** *Brown et al. 2025; Li et al. 2022; Bassi et al. 2024*
Limitation	**Internal spatial support removed.** Static and dynamic information combined by simple fusion.	**Distributed variables stay heterogeneous:** inconsistent resolution and projection across sources.	**Not hydrology-aware:** general representations ignore processes across basins, tasks and regimes.
Gap	1 **Distributed models that preserve basin structure and fuse dynamic forcing with static variables better**	2 **A standardized, basin-aligned dataset for large-scale distributed hydrologic learning**	3 **Basin representations that keep hydrologic information and transfer across tasks, regions and climates**

Each gap sets up one of the studies that follow.

Selected Publications

Title	Venue	Completion Date	Status
Watershed-Agnostic Forecasting: An Explainable, Distributed, Data-Driven Hydrologic Model	**Environmental Modelling & Software**	**Jun 2025**	**Published**
From Aggregated to Distributed: A Transformer Framework for Generalizable Streamflow Modeling Across Space and Time	**Engineering Applications of AI**	**Dec 2025**	**Revisions Submitted**
BASINGRID: A distributed 40-year spatiotemporal hydrology dataset for U.S. basins	**Nature Scientific Data**	**Apr 2026**	**Published**
Basin2Vec: Learning Basin-Centric Embeddings for Scalable Hydrological Modeling	**CVF WACV 2027**	**Jun 2026**	**Accepted**

STUDY 1

Fusion of Static and Dynamic Variables

HydroTransformer · the basin as a spatial field

Study 1: Introduction

Typical Lumped Modeling

Dynamic variables X_t (basin averages)

P	Tmax	Tmin	SRad	VP	SWE

⋮ one value per day

Static attributes S

Elev	Slope	Clay	Sand	AWC	FC

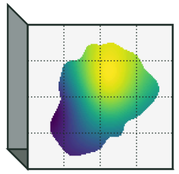
[X_t ; S]
stacked as
extra channels

LSTM → *Q(t)*

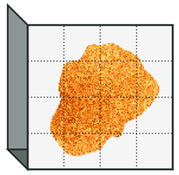
Spatial organization is lost before the model sees the data.

Distributed Basin-Aware Modeling

Dynamic variables X_t (gridded)



Static variables S



Spatial encoder

✗ **No channel stacking**

Conditional fusion
static context
gates dynamic
features

→ *Q(t)*

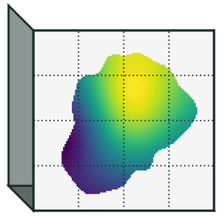
Where things happen inside the basin is kept.

Study 1: condition distributed dynamic features on distributed static basin structure.

Study 1: Overview

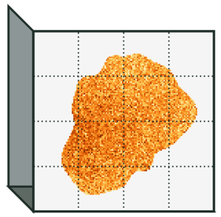
QUESTION Which variables are used to predict streamflow across time and in unseen basins?

Dynamic Variables



1. Precipitation
2. Min and max temperature
3. Shortwave radiation
4. Vapor pressure
5. Daylength
6. Snow water equivalent

Static Variables



1. Elevation
2. Clay, sand, and silt
3. Available water capacity
4. Field capacity

Evaluation Settings

Temporal

future periods in observed basins

Spatial

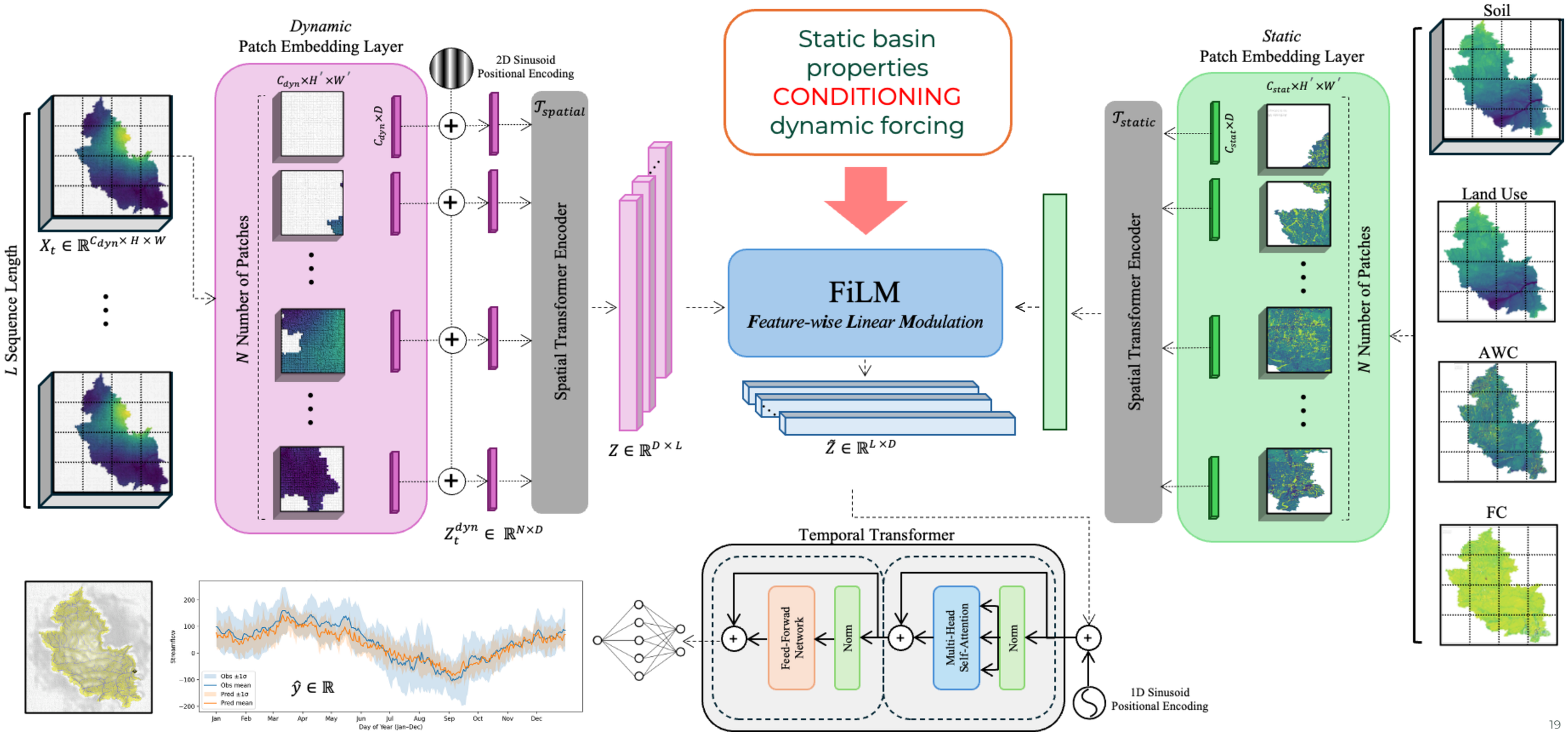
completely held-out basins

$$\hat{Q}_{b,t} = \mathrm{HydroTransformer}\left(X_{b,\, t-L+1:t},\ S_b\right)$$

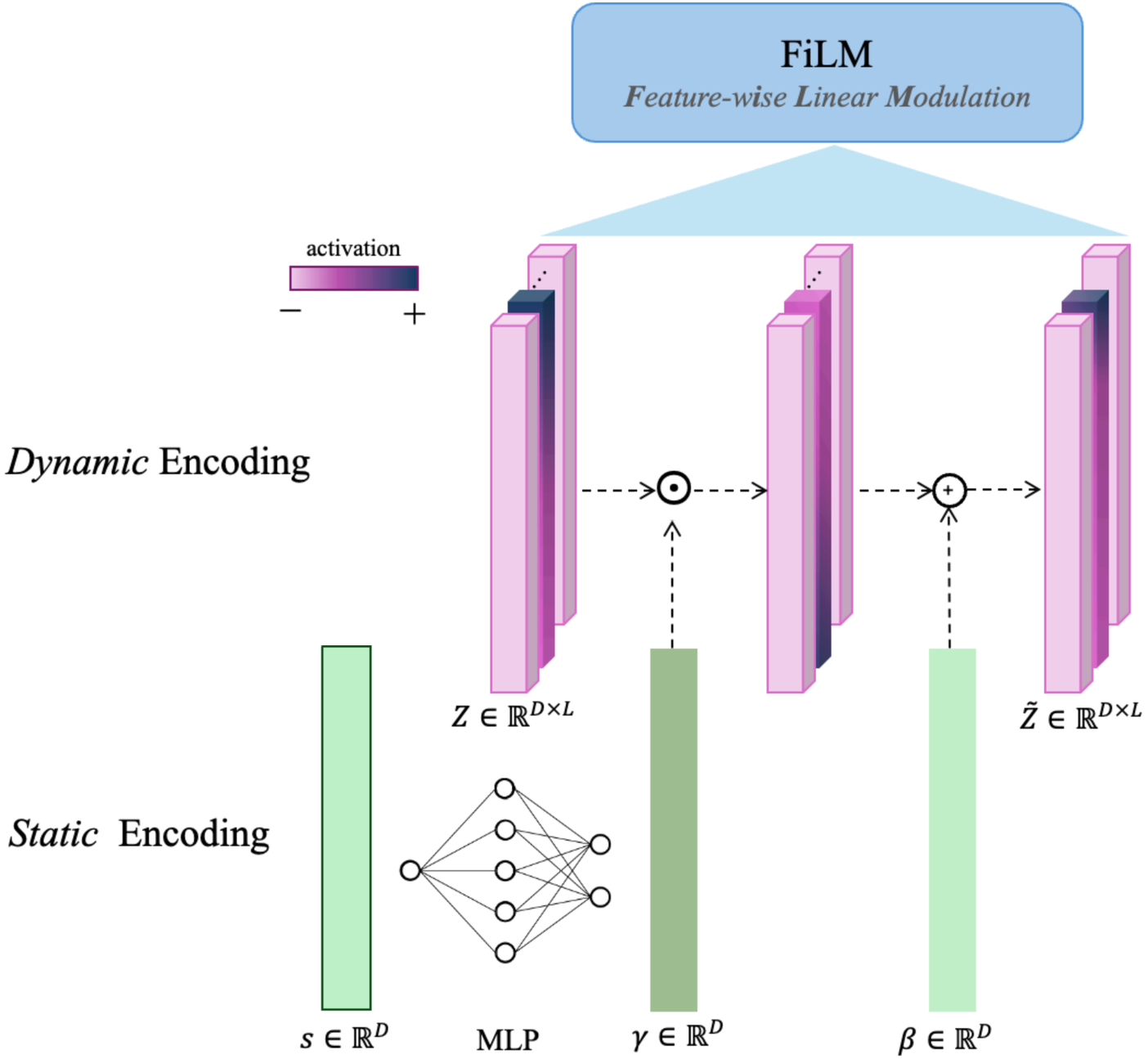
$X_{b,\, t-L+1:t}$	gridded dynamic variables over the last L days
S_b	gridded static variables of basin b
$\hat{Q}_{b,t}$	predicted discharge for basin b on day t

Study 1: HydroTransformer

Encode dynamic and static basin information separately, then integrate them through conditioning.



Study 1: FiLM Conditioning



How static variables are CONDITIONING dynamic variables?

$$\tilde{z}_{b,t} = z_{b,t} \odot (1 + \gamma_b) + \beta_b$$

γ_b: Basin-specific **feature scaling**

β_b: Basin-specific **feature shifting**

Study 1: Experimental setup

Does distributed basin structure improve generalization across time and unseen basins?

Evaluation Settings

Temporal generalization: future periods in observed basins

Spatial generalization: completely held-out basins

Benchmarks

LSTM · EA-LSTM · Informer · CNN-LSTM · HydroTransformer

Generalization Tests

Sequence Lengths: {5, 21, 365}

Temporal generalization

Train: 2000–2013

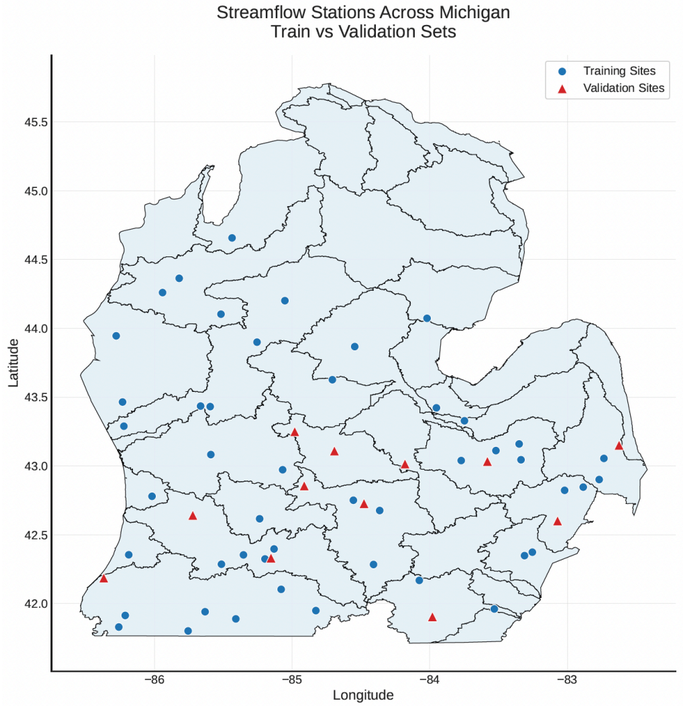
Test: 2014–2020

Same basins, future period

Spatial generalization

12 completely held-out basins

No discharge observations from test basins used during training



Evaluation Metrics

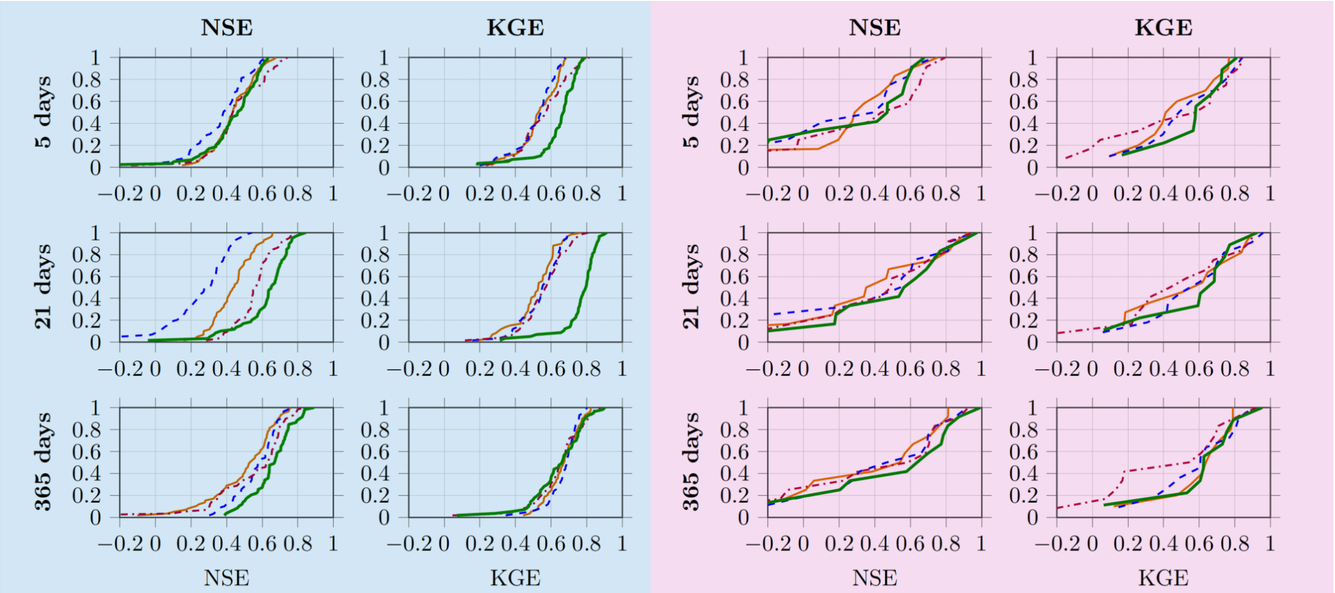
Nash–Sutcliffe Efficiency (NSE)
Range: (−∞, 1]

Kling–Gupta Efficiency (KGE)
Range: (−∞, 1]

Study 1: Generalization Performance

With these two evaluation settings fixed, does distributed basin structure actually improve prediction?

CDF Plots



(a) Temporal generalization performance distributions.

(b) Ungauged-basin generalization performance distributions.

LSTM EA–LSTM Informer HydroTransformer

NSE Evaluation

Temporal Generalization

	No static			With static		
Model	5	21	365	5	21	365
LSTM	0.180	0.063	0.119	0.439	0.444	0.535
EA-LSTM	–	–	–	0.383	0.309	0.579
CNN–LSTM	0.363	0.529	**0.639**	–	–	–
Informer	0.242	0.300	0.370	0.436	0.561	0.628
HydroTransformer	**0.458**	**0.658**	0.445	**0.463**	**0.659**	**0.670**

Spatial Generalization

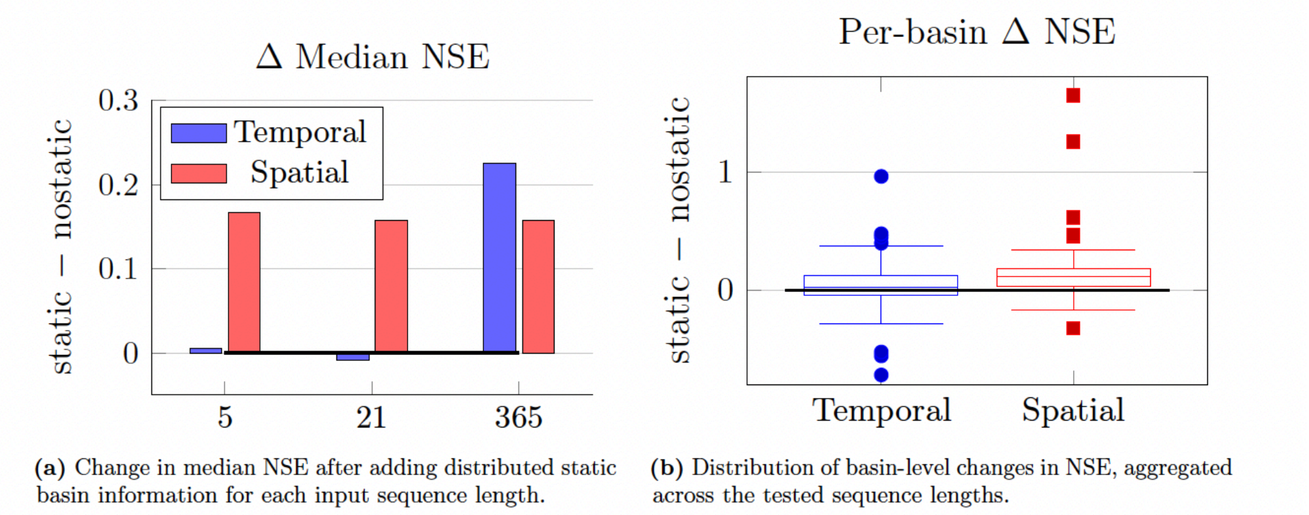
	No static			With static		
Model	5	21	365	5	21	365
LSTM	0.273	0.455	0.590	0.314	0.408	0.555
EA-LSTM	–	–	–	0.420	0.547	0.563
CNN–LSTM	0.295	0.444	0.595	–	–	–
Informer	**0.574**	**0.554**	**0.609**	**0.512**	0.493	0.634
HydroTransformer	0.304	0.439	0.510	0.470	**0.596**	**0.668**

HydroTransformer improves generalization, most clearly in ungauged basins.

Study 1: Impact of Distributed Static Attributes

How does static basin information improve HydroTransformer's performance?

Performance Gain



(a) Change in median NSE after adding distributed static basin information for each input sequence length.

(b) Distribution of basin-level changes in NSE, aggregated across the tested sequence lengths.

$$\Delta\text{NSE} = \text{NSE}_{\textit{with static}} - \text{NSE}_{\textit{without static}}$$

Temporal

$$\Delta\text{NSE} = \begin{cases} +0.005, & 5d \\ +0.001, & 21d \\ +0.225, & 365d \end{cases}$$

Spatial

$$\Delta\text{NSE} = \begin{cases} +0.166, & 5d \\ +0.157, & 21d \\ +0.158, & 365d \end{cases}$$

Distributed static variables improve generalization.

Study 1: Conclusion

Study 1 shows that keeping the basin as a spatial field improves generalization.

Established

- **Distributed fields** carry more information than basin-averaged (lumped) inputs.
- **Conditioning** dynamic fields on static basin structure outperforms simple concatenation.
- The gains are largest in **unseen basins**.

Next Challenge

- The workflow is region-specific: it was built for 61 basins in Michigan.
- Distributed rasters are computationally expensive to process, which limited us to six static variables.
- Scaling to thousands of basins requires standardized data.

Next: BASINGRID, a standardized dataset

STUDY 2

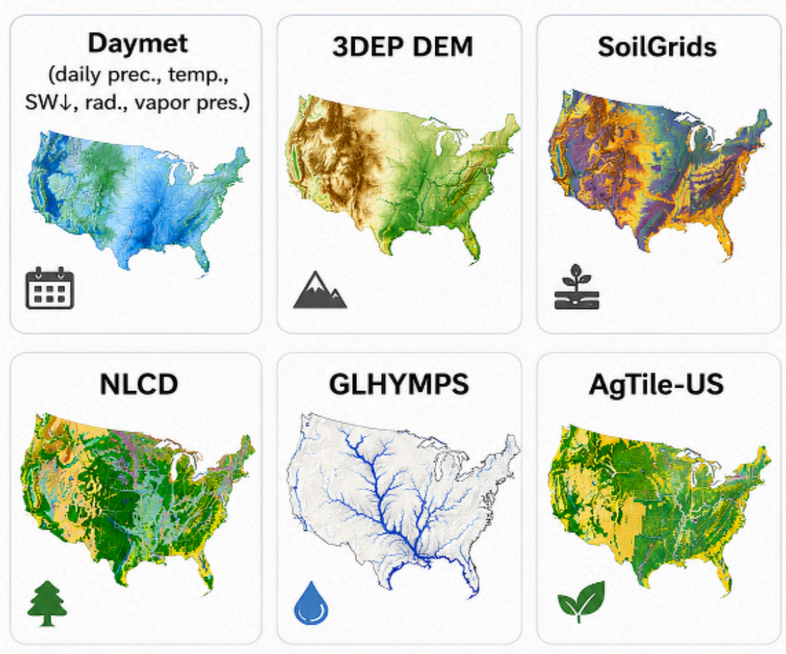
BASINGRID

A distributed 40-year spatiotemporal hydrology dataset for U.S. basins

Study 2: Motivation

Trade-off of scale and spatial structure

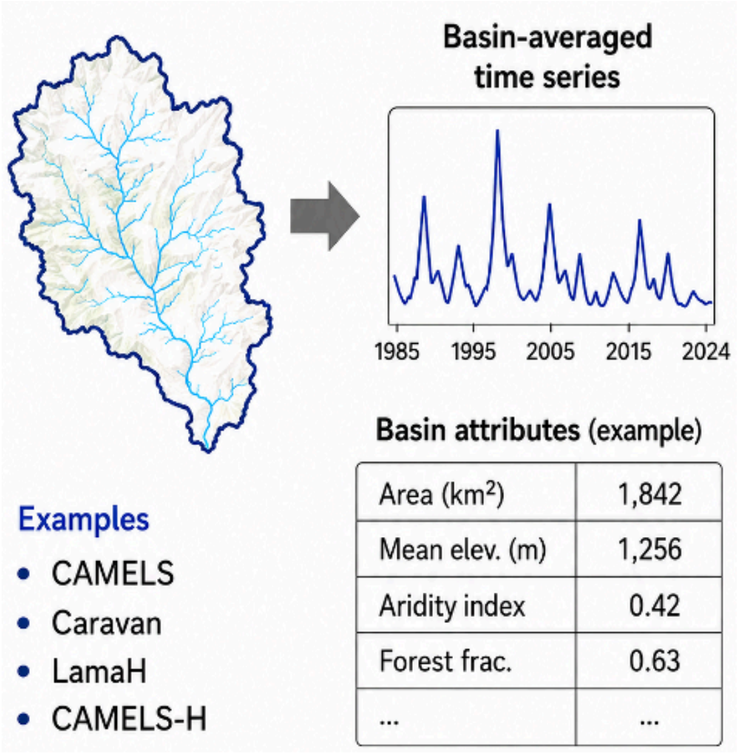
Raw gridded data sources



Different projections, resolutions and coverage

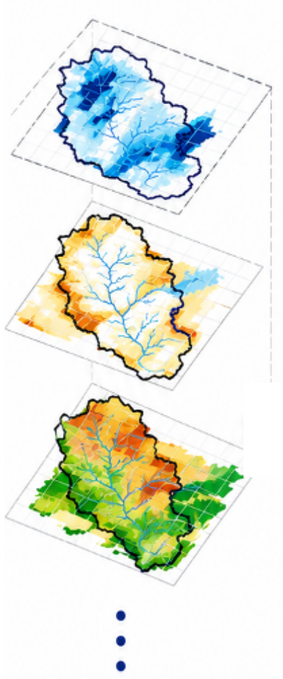
Heterogeneity

Existing large-scale datasets



Loss of spatial structure

What do we need?



Preserve spatial information and make the data scalable

Study 2: Variables and data sources

16 variables at three native temporal resolutions

DAILY

365 time steps / year
6 variables

Meteorological forcings

Precipitation · minimum and maximum temperature · shortwave radiation · snow water equivalent · vapor pressure

SOURCE

Daymet: ~1 km gridded daily meteorology

ANNUAL

1 time step / year
2 variables

Land-surface evolution

Potential evapotranspiration (PET) · land-cover class

PET is computed daily with Hargreaves and stored as an annual sum.

SOURCES

PET: derived from CW3E
Land cover: NLCD

STATIC

no time dimension
8 variables

Basin attributes

Topography: elevation · **Soil:** clay, silt, sand, bulk density
Hydrogeology: permeability, hydraulic conductivity ·
Agriculture: tile drainage

SOURCES

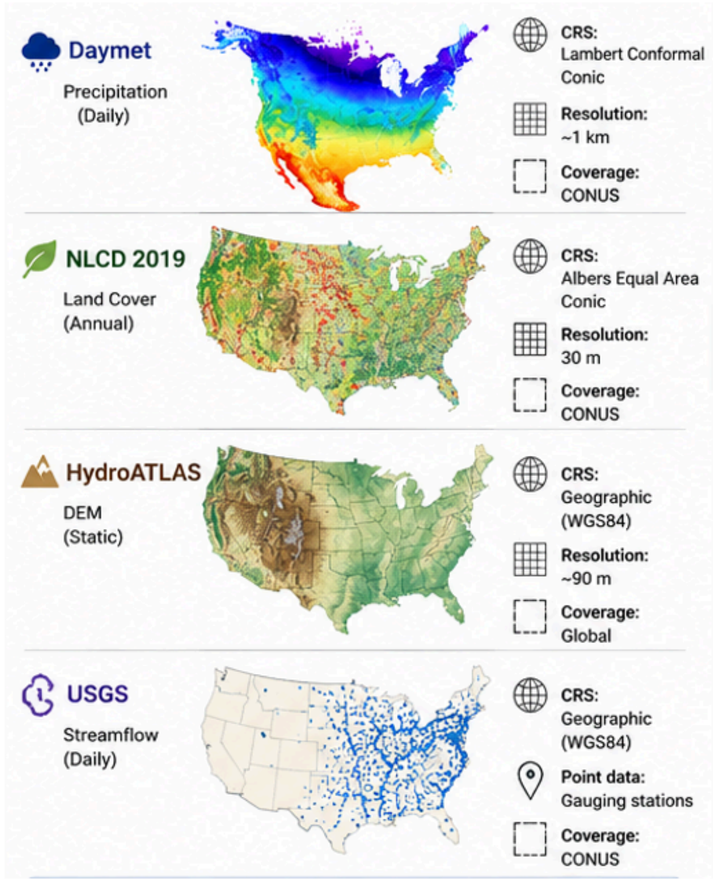
DEM: USGS 3DEP · Soil: SoilGrids
Subsurface: GLHYMPS · Tile drainage: AgTile-US

Different coordinate reference systems + different spatial and temporal resolutions

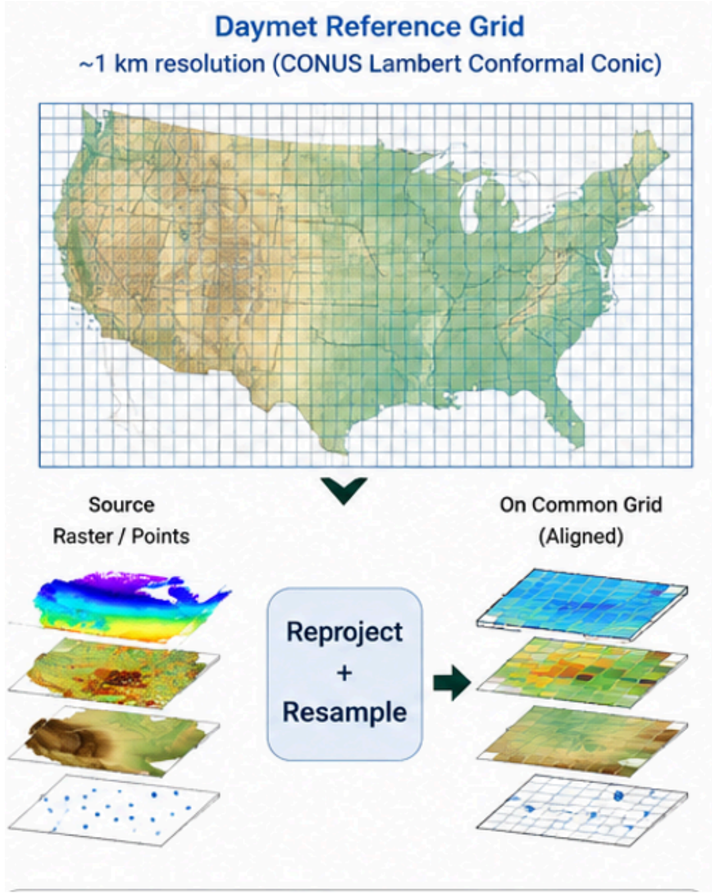
Study 2: A common grid for every basin

Different sources (CRS, resolution, coverage) ⟶ **one reference grid** ⟶ **basin-aligned 64 × 64**

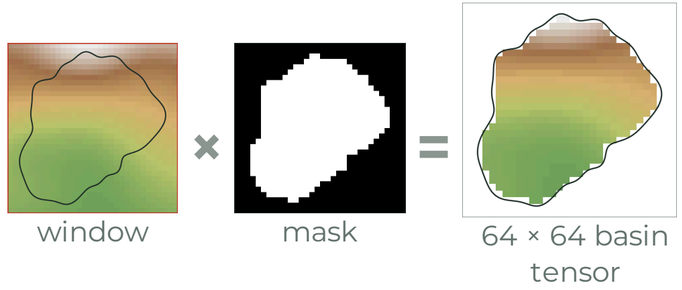
Heterogeneous source grids



Common grid



Basin-aligned output



values + mask + physical-scale metadata

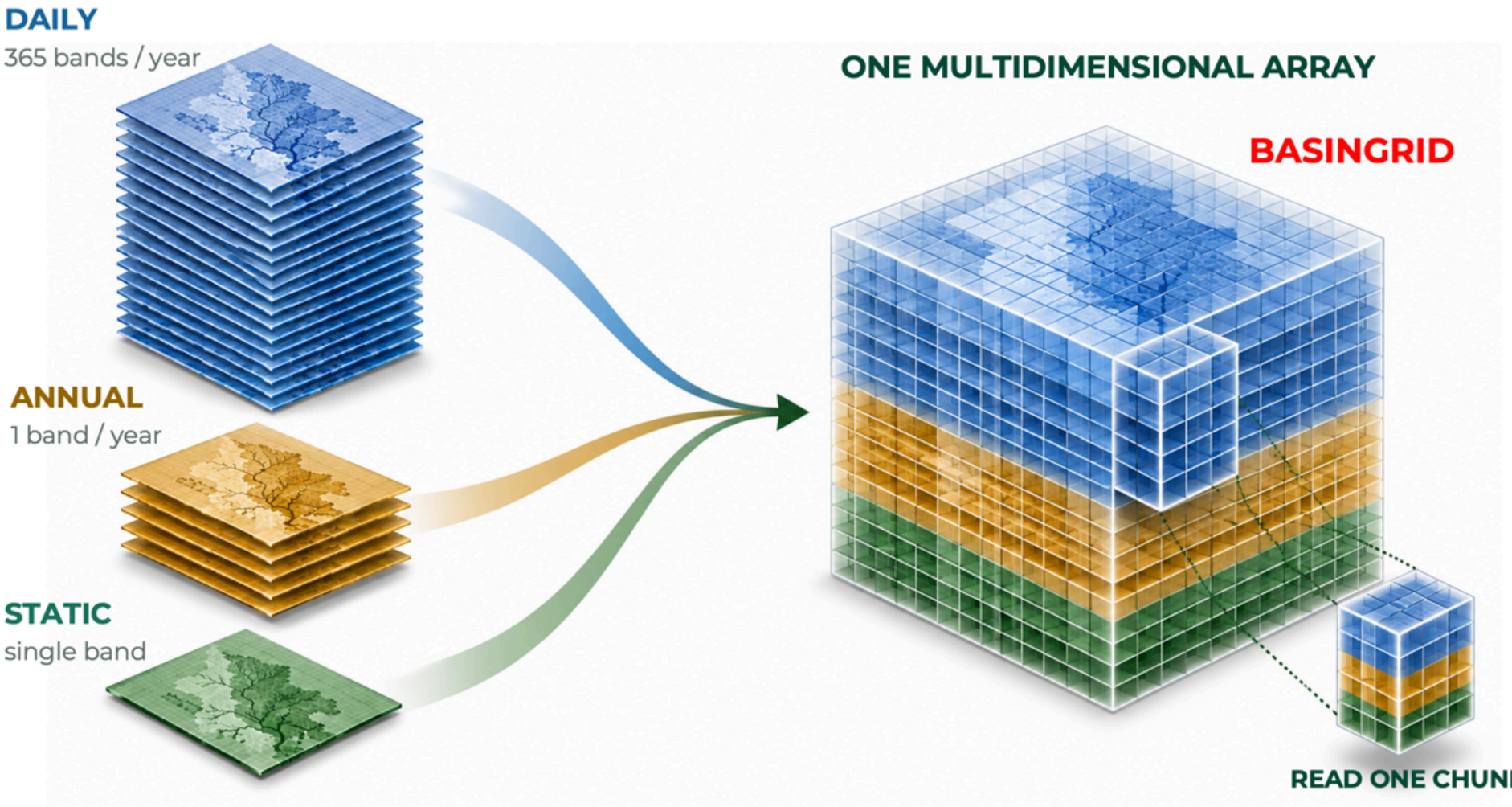
Continuous variables: bilinear interpolation

Categorical variables: nearest neighbor

Standardized CRS · aligned grid · consistent tensor shape

Study 2: Organizing the Data Cube

Native temporal resolutions are preserved within one model-ready array structure.



ZARR STORAGE

Designed for selective, large-scale access

- **CHUNKED**
  Basin and temporal blocks are accessed independently.
- **COMPRESSED**
  Blosc with Zstandard reduces storage demand.
- **SCALABLE**
  Parallel workers and resumable execution.

ARRAY INDEX $BASIN \times YEAR \times TIME \times VARIABLE \times 64 \times 64$

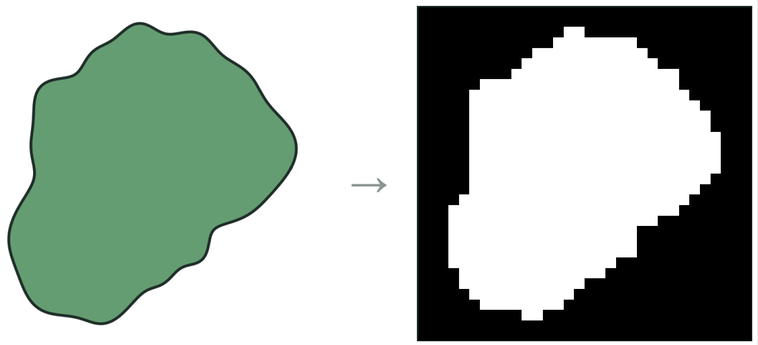
PRESERVE TEMPORAL RESOLUTION · STANDARDIZE SPATIAL SHAPE · SCALE ACCESS ACROSS 9,067 BASINS

Study 2: Validation metrics

Three complementary checks: **basin area**, **continuous values**, and **categories**

Area preservation

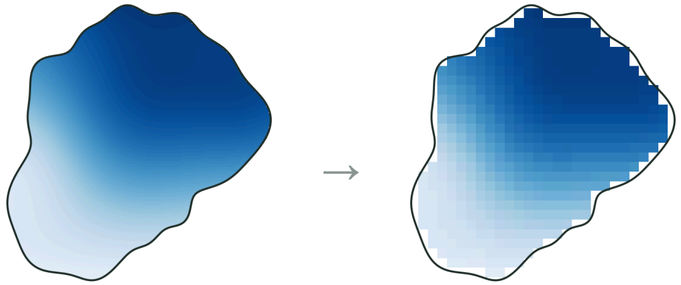
Does the normalized mask keep the basin area?



$R_A = A_{\text{mask}} / A_{\text{vector}}$
Ideal: $R_A = 1$, same physical area

Continuous values

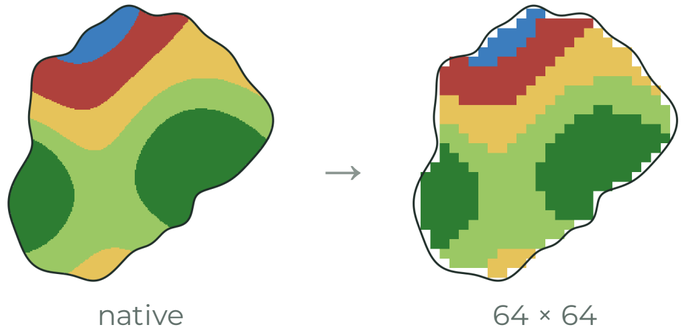
Does the patch keep continuous values?



MdAPE: typical % difference from the native raster
Ideal: MdAPE → 0%

Categorical fidelity

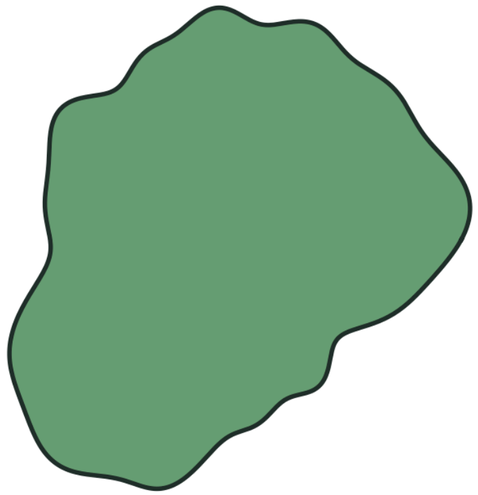
Are land-cover categories kept?



Class-proportion distance and **dominant-class agreement**
Ideal: 0% distance, 100% agreement

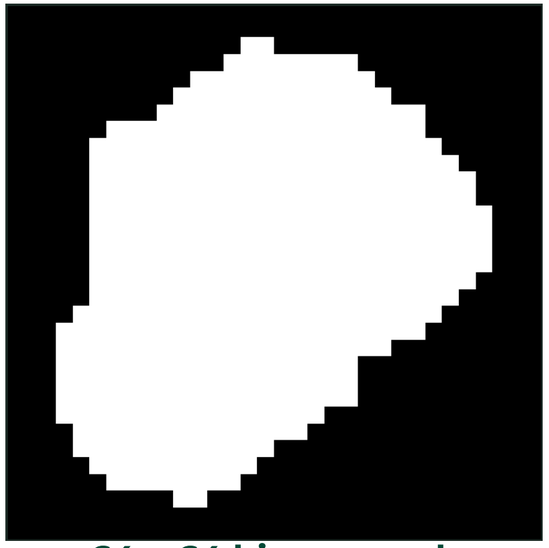
Study 2: Validation, basin area is preserved

After restoring basin-specific pixel area, the normalized mask closely matches the original vector basin.



Original vector basin

→



64 × 64 binary mask

Area agreement · 9,067 basins

1.094

median mask / vector area ratio

1.100

mean mask / vector area ratio

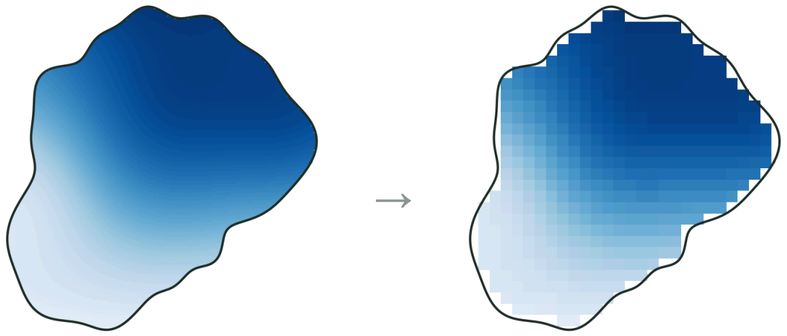
Normalization changes the **effective resolution**, not the basin's **physical area**.

The 64 × 64 normalization keeps basin-size information.

Study 2: Validation, resampled values

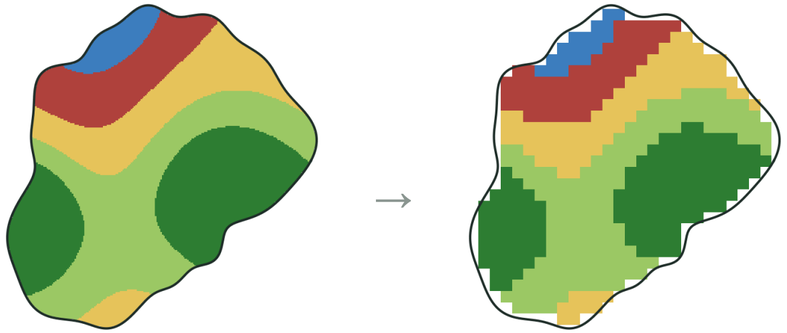
Native-grid and 64 × 64 patch values stay closely aligned across small, medium, and large basins.

Continuous: precipitation



native grid → 64 × 64 patch

Categorical: land cover



native grid → 64 × 64 patch

Precipitation fidelity

< 1%

MdAPE in every basin-size class

Land-cover fidelity

100%

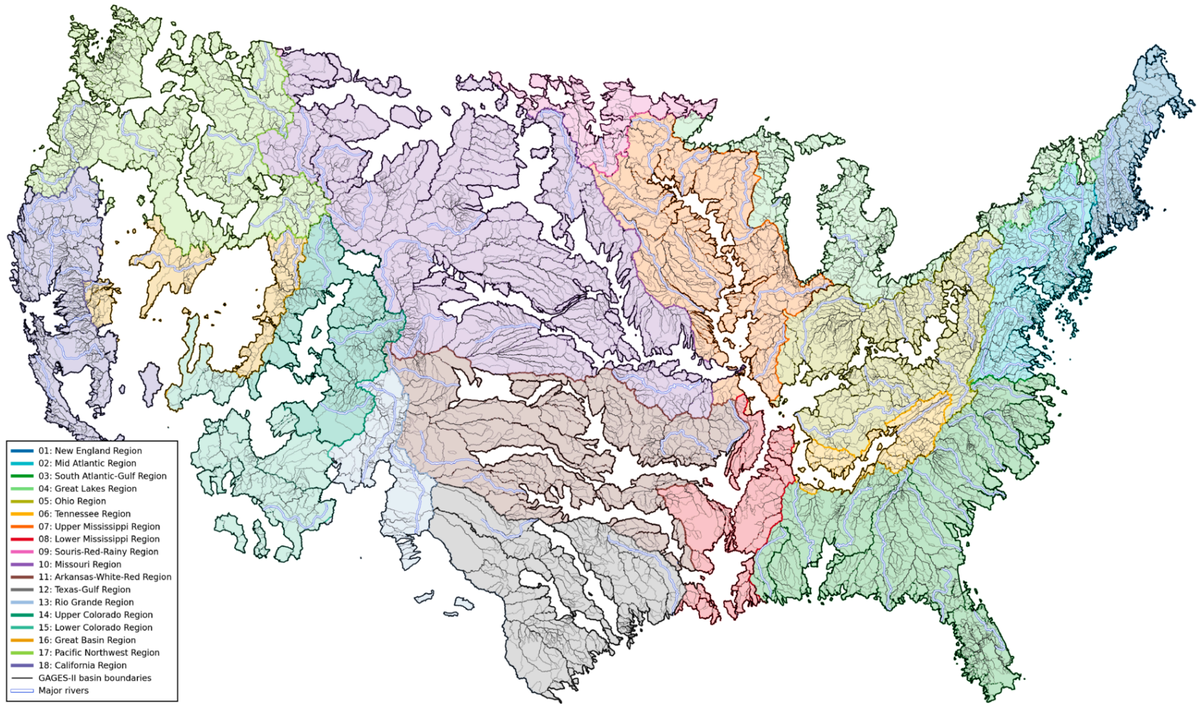
dominant-class agreement

class-proportion distance 1.23–2.89%

The 64 × 64 representation keeps basin-scale information across basin sizes.

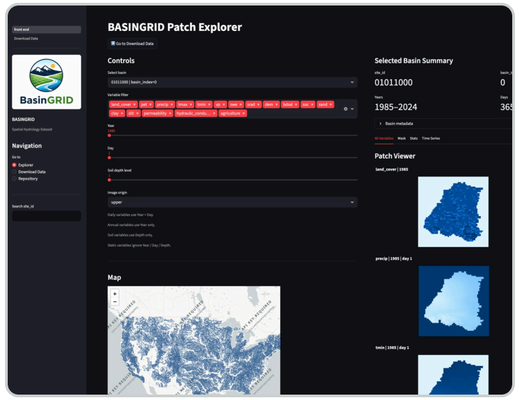
Study 2: BASINGRID

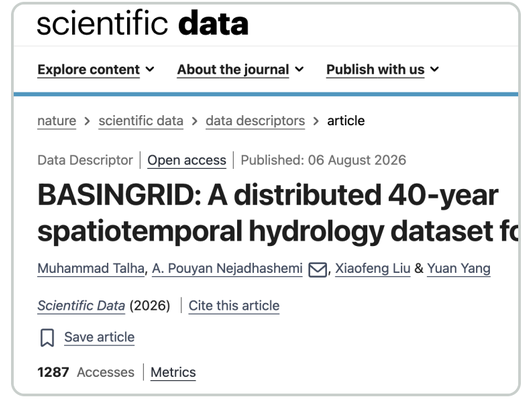
Spatial coverage



Dataset

Spatial scale	**9,067** GAGES-II basins, CONUS
Time period	**1985–2024**
Resolution	Daily · 64 × 64 per basin
Format	ML-ready Zarr chunks
Variables	16 · 6 daily, 2 annual, 8 static





basingrid.ciwre.msu.edu · Scientific Data

BASINGRID turns distributed basin modeling from a study-specific workflow into reusable infrastructure.

Study 2: Conclusion

Study 2 shows that the spatial-field representation can be standardized and scaled to the national level.

Established

- Heterogeneous rasters can be harmonized into **common basin-aligned tensors**.
- The 64 × 64 patches preserve basin **geometry, values, and categories**.
- The pipeline scales to **9,067 basins** over **1985–2024**, using chunked, parallel storage.

Next Challenge

- Each basin is now a stack of variables × time × 64 × 64 values.
- Every downstream model would have to learn to read this stack from scratch.
- The tensors are **too high-dimensional** to reuse directly.

Next: Basin2Vec, a compact basin vector

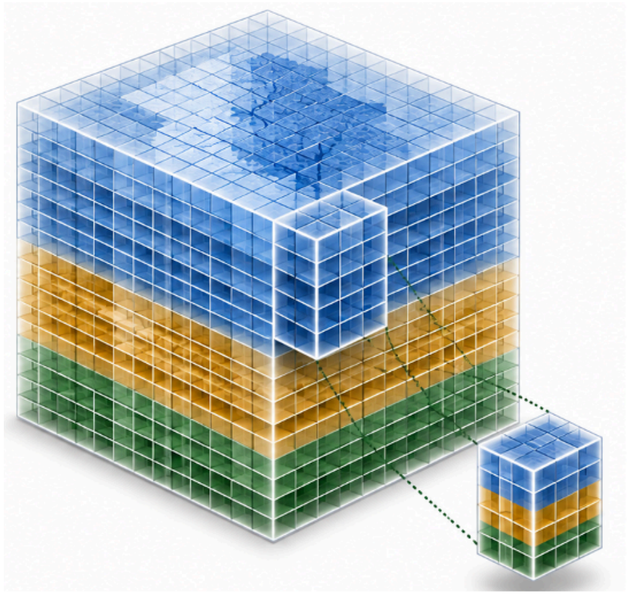
STUDY 3

Basin2Vec

Learning basin-centric embeddings for scalable hydrologic modeling

Study 3: Motivation

What we have now



BASINGRID: dynamic static annual basin mask

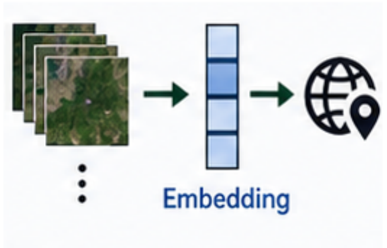
Spatially rich, but high-dimensional for downstream models

Current compact alternatives

f_1	f_2	f_3	⋯	f_p
0.82	-1.34	0.57	⋯	1.11

Feature vector

A. CAMELS attributes

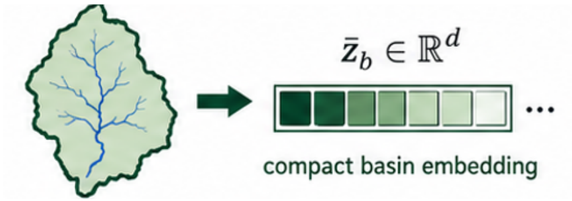


B. EO embeddings

A: loses spatial structure

B: not hydrology-aware

What do we need?



One compact vector per basin

Learned embeddings from hydrology data

Study 3: Why existing representations fall short

Comparison

Approach	Basin-indexed	Spatial & scale-aware	Hydrologic relation-aware
CAMELS attributes	●●●	●●●	●●●
AlphaEarth	●●●	●●●	●●●
TESSERA	●●●	●●●	●●●
SatCLIP	●●●	●●●	●●●
Basin2Vec (ours)	●●●	●●●	●●●

●●● High ●●● Moderate ●●● Low

Requirement 1

Basin-indexed

One representation for one basin.

Requirement 2

Spatial and scale-aware

Learns within-basin structure and keeps physical basin size.

Requirement 3

Hydrologic relation-aware

Embedding geometry reflects hydrologic relations.

Goal: a compact, basin-indexed representation whose geometry is organized by hydrologic relations.

Study 3: Methodology overview

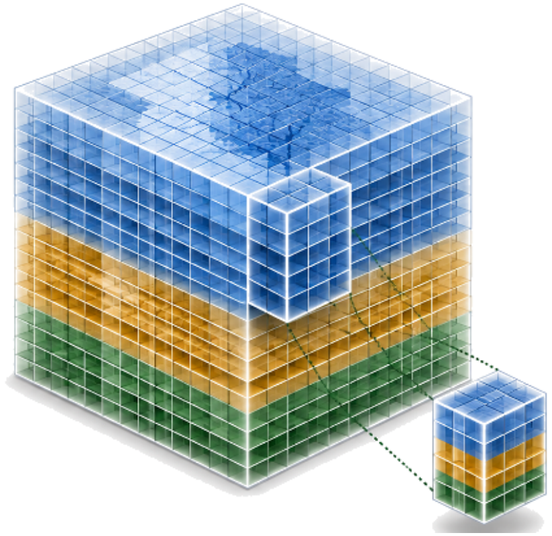
From a **basin-year tensor** to a **reusable basin embedding**

Input entity

(b, y)

one basin-year

Basin-year tensor



Monthly dynamic · **annual** · **static**

Basin mask + **metadata**, from BASINGRID

Basin2Vec

Scale-aware encoder

mask-aware spatial encoding of the tensor

Relation-aware self-supervision

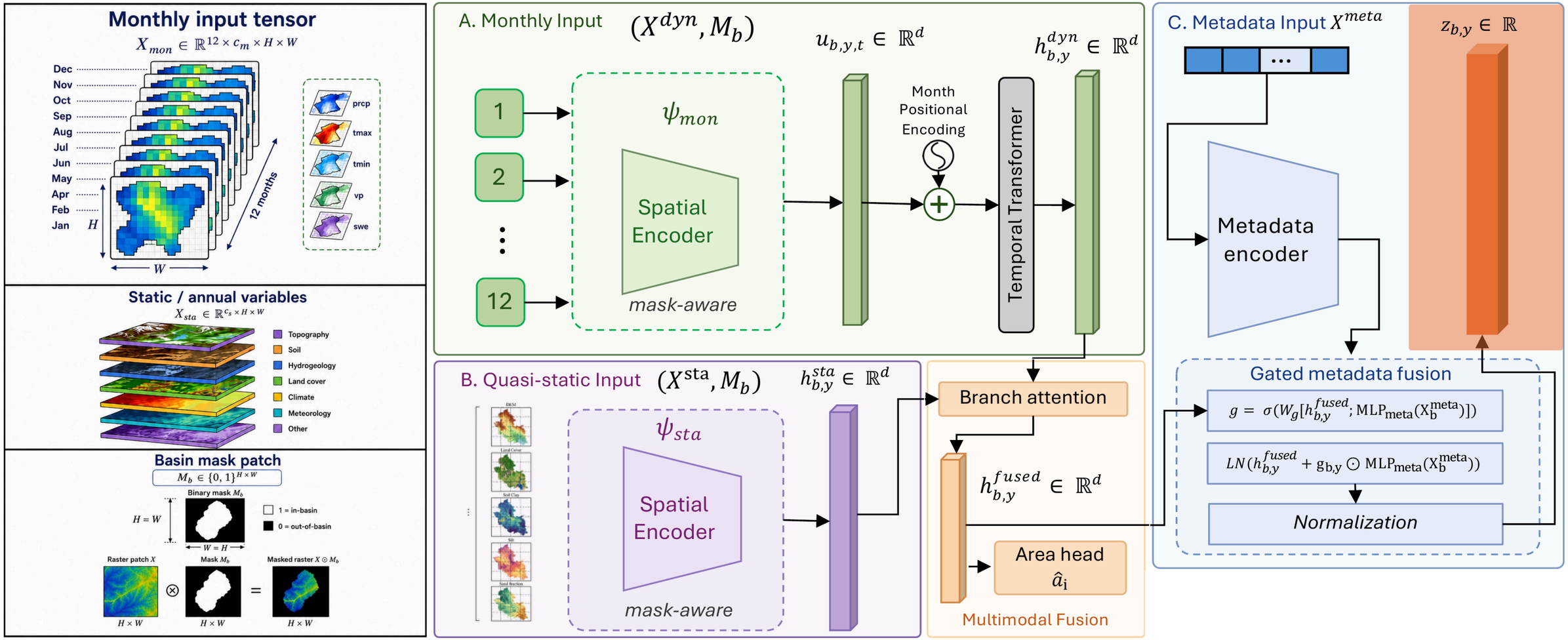
hydrologic relations shape the embedding space

No observed streamflow is used

Output

$z_{b,y} \in \mathbb{R}^d$

Study 3: Scale-aware encoder



Mask-aware

Out-of-basin cells never enter the pooling.

Area head

Predicts drainage area, so physical basin size is kept.

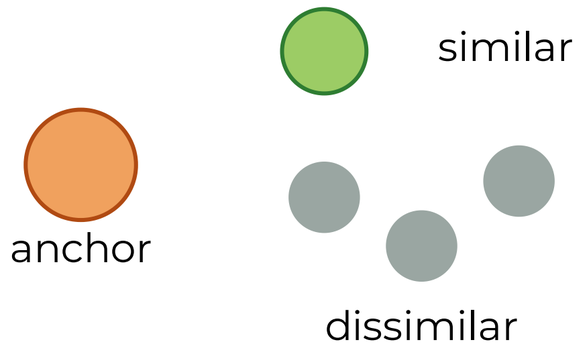
Gated metadata

Basin geometry is added through a learned gate.

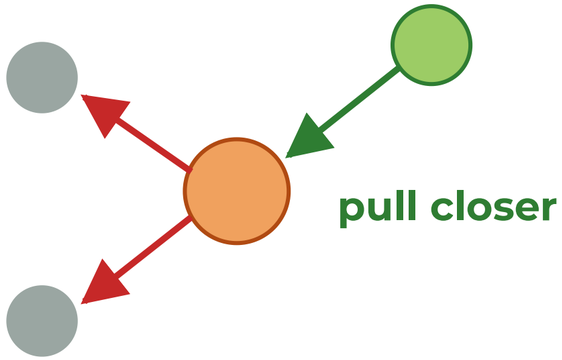
Study 3: Contrastive learning

Learn an embedding space by pulling related samples together and pushing unrelated samples apart.

Original and candidate samples

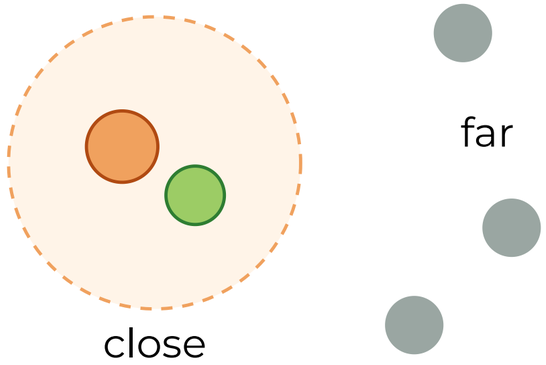


Contrastive learning rule



maximize similarity to positives,
minimize similarity to negatives

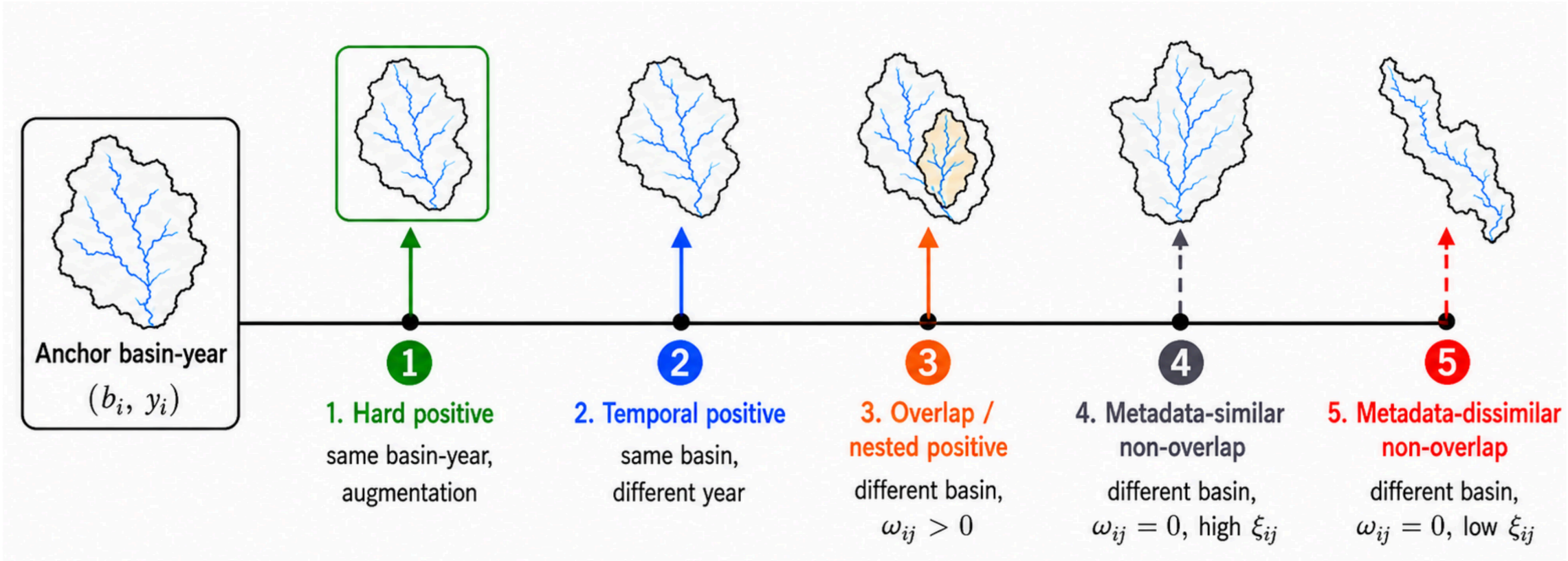
Learned embedding space



The standard rule treats only two views of the same sample as related. We extend it to **hydrologically meaningful relations** between basins.

Study 3: Hydrology-aware relations

How do we define hydrologically meaningful relations among basin-years?



Temporal positive

weight decays with the gap in years, $e^{-|y_i - y_j|/\tau_y}$

Overlap / nested positive

weighted by overlap, $\omega_{ij} = \frac{|G_i \cap G_j|}{\min(|G_i|, |G_j|)}$

Negatives

pushed apart more weakly when basin metadata are similar

We use **time**, **space**, and **basin similarity** to structure the embedding space.

Study 3: Hydrology-aware contrastive loss

$$\mathcal{L} = \mathcal{L}_{con} + \lambda_n \mathcal{L}_{neg} + \lambda_a \mathcal{L}_{area}$$

Relation attraction

$$\mathcal{L}_{con}$$

- **Pull** hard, temporal, and overlap / nested positives toward the anchor
- Strength set by the relation weights

+

Negative separation

$$\mathcal{L}_{neg}$$

- **Push** non-overlapping basins below a similarity margin
- Margin is looser for metadata-similar basins

+

Scale preservation

$$\mathcal{L}_{area}$$

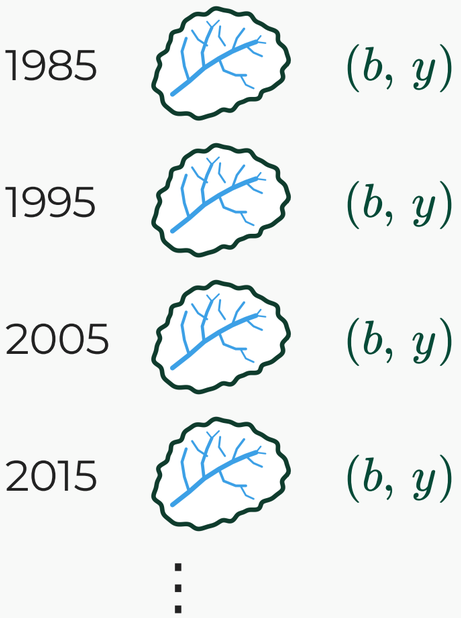
- **Keep** basin size: predict standardized log drainage area
- Smooth-L1, robust to very large basins

Pull related basin-years together, push unrelated basins apart, and keep basin scale.

Study 3: Final product

From basin-year embeddings to one reusable basin descriptor

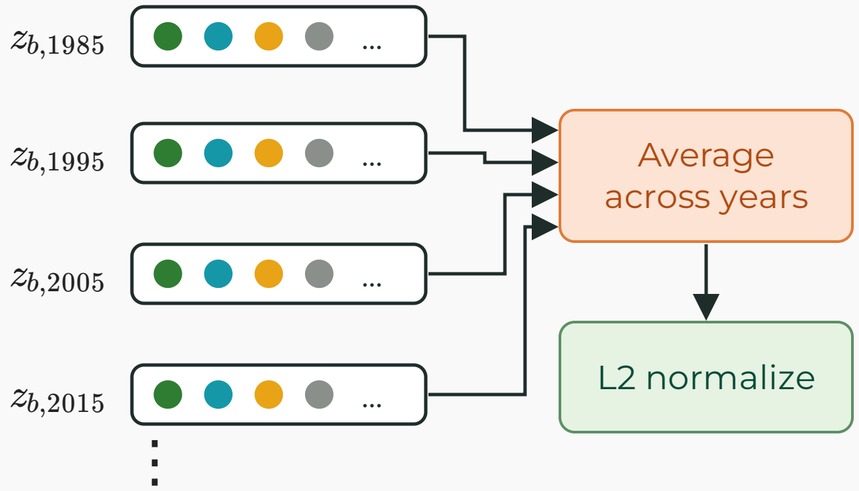
A basin across years



same basin, different years

Encode each basin-year

pretrained Basin2Vec encoder

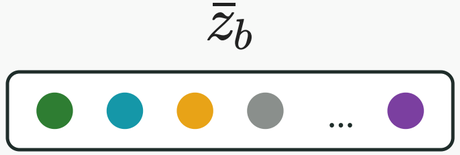


$z_{b,y} \in \mathbb{R}^{64}$

one 64-D embedding per basin-year

$$\bar{z}_b = \text{Normalize}\left(\frac{1}{|Y_b|}\sum_{y \in Y_b} z_{b,y}\right)$$

Final Descriptor



fixed basin descriptor

one reusable 64-D descriptor per basin

One fixed descriptor is used for the downstream streamflow prediction task.

Study 3: Downstream evaluation setup

Same prediction setup; only the basin descriptor changes.

$$\hat{Q}_{b,t} = F_\eta\big([\, x^{met}_{b,t-L+1}, \ldots, x^{met}_{b,t} \,],\ c_b\big), \qquad L \in \{120,\ 365\}$$

Daily weather x^{met}

Daymet precipitation, shortwave radiation, max and min temperature, vapor pressure

no observed streamflow as input

Basin descriptor c_b · frozen

Basin2Vec

- CAMELS attributes (27)
- AlphaEarth
- TESSERA
- SatCLIP

Predictor F_η

- LSTM
- TFT
- TimeXer
- Timer-XL
- iTransformer

Any difference in skill comes from the **representation**, not the model.

Study 3: Generalization experiments

Does the same descriptor generalize across time and across space?

Temporal generalization

671 CAMELS-US basins, same basins in both periods

Zero-shot spatial transfer

6-fold ungauged-basin cross-validation

- Held-out basins excluded from training and early stopping
- Train 1990–2004, validate 2005–2014 on the rest
- Held-out basins evaluated over 1990–2014

Evaluation: median NSE ↑ and KGE ↑ across basins · 3 seeds

Study 3: Temporal generalization results

Median NSE · future period · L = 365

Predictor	Basin2Vec	CAMELS attr.	AlphaEarth	TESSERA	SatCLIP
LSTM	**0.782**	0.747	0.641	0.662	0.719
TFT	**0.753**	0.717	0.741	0.570	0.638
TimeXer	**0.651**	0.502	0.630	0.573	0.603
Timer-XL	**0.730**	0.591	0.674	0.627	0.402
iTransformer	**0.627**	0.555	0.559	0.576	0.464

Mean of three seeds. 671 basins, test 2010–2014. Basin2Vec is best in every row.

Average relative gain

up to 8.3%

over the best alternative

Consistency

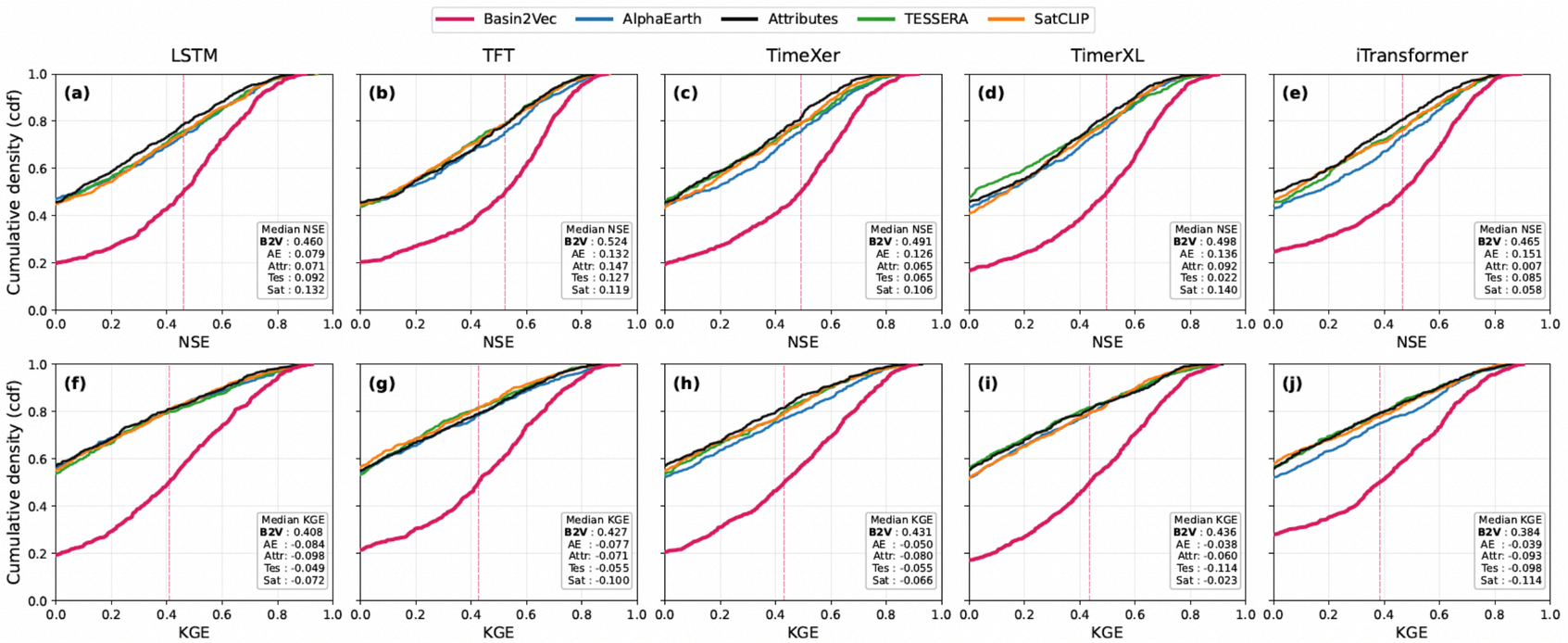
5 / 5

predictors: first in NSE and KGE, at L = 120 and 365

Basin2Vec consistently improves future-period prediction across predictors and window lengths.

Study 3: Zero-shot spatial transfer results

NSE and KGE CDFs · held-out basins



Median NSE · held-out basins · L = 365

Median NSE	LSTM	TFT	TimeXer	Timer-XL	iTransformer
Basin2Vec	**0.460**	**0.524**	**0.491**	**0.498**	**0.465**
CAMELS attr.	0.071	0.147	0.065	0.092	0.007
AlphaEarth	0.079	0.132	0.126	0.136	0.151
TESSERA	0.092	0.127	0.065	0.022	0.085
SatCLIP	0.132	0.119	0.106	0.140	0.058

Basin2Vec reaches **3.5×** the median skill of the strongest baseline on unseen basins.

Study 3: Ablation study

Median NSE · LSTM · L = 365

Variant	Temporal	Spatial
Full Basin2Vec	**0.782**	**0.460**
Scale-aware encoder		
w/o monthly branch	0.649	0.423
w/o quasi-static branch	0.635	0.404
w/o metadata conditioning	0.634	**0.064**
w/o area regularization	**0.601**	0.412
Relation-aware objective		
hard positives only	0.670	**0.372**
w/o temporal positives	0.621	0.418
w/o overlap / nesting positives	0.633	0.408
w/o negative-margin loss	0.643	0.409

Encoder

Removing **metadata conditioning** collapses unseen-basin skill: **0.460** → **0.064**. The **area term** matters most over time.

Objective

Standard contrastive learning (hard positives only) transfers worst: **0.372** on unseen basins.

Every component helps; the full model is best on every metric.

Study 3: Conclusion

Study 3 shows that learned basin embeddings outperform the other representation options, especially in unseen basins.

Established

- **Learned embeddings beat** CAMELS attributes and Earth-observation embeddings.
- First for **all 5 predictors**: up to **8.3%** gain over time, **3.5×** median NSE in unseen basins.
- **Every component helps**, most of all metadata for unseen-basin transfer.

Next Challenge

- One descriptor per basin: one outlet, nothing about the reaches inside.
- Ungauged reaches need the river network explicitly.

Next: the basin as a directed river network

STUDY 4 · ONGOING

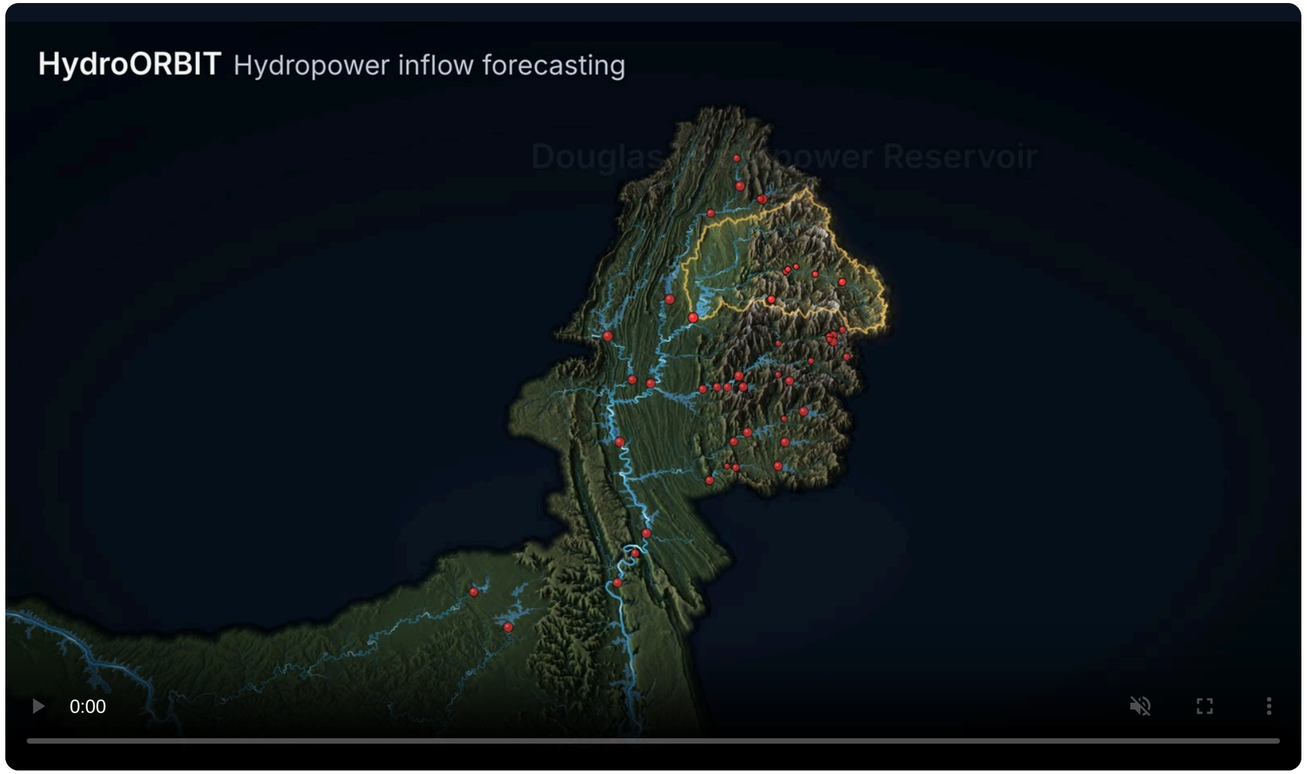
A River-Network GNN for the Tennessee River Basin

The basin as a directed river network

Study 4: Motivation

MOTIVATION Water managers need forecasts where there is no gauge.

Example: forecasting inflow to a hydropower reservoir



1 · The need

Managers need forecasts **days ahead**, also on rivers **with no gauge**.

2 · Prior work

GNNs on the Danube: **river topology added almost nothing**.

Kirschstein & Sun, 2024 · every node gauged, 6 h ahead

3 · Our setting

Most reaches are **ungauged** and leads reach **10 days**: the network must **carry skill** where no data exist.

Question: can a river network carry forecast skill to reaches it has never observed?

Study 4: Problem, forecasting where there is no gauge

QUESTION THIS ANSWERS Can one model forecast flow at every reach, and keep its skill at gauges it never saw?

58,834

NHDPlus reaches in the Tennessee River Basin (~106,000 km²)

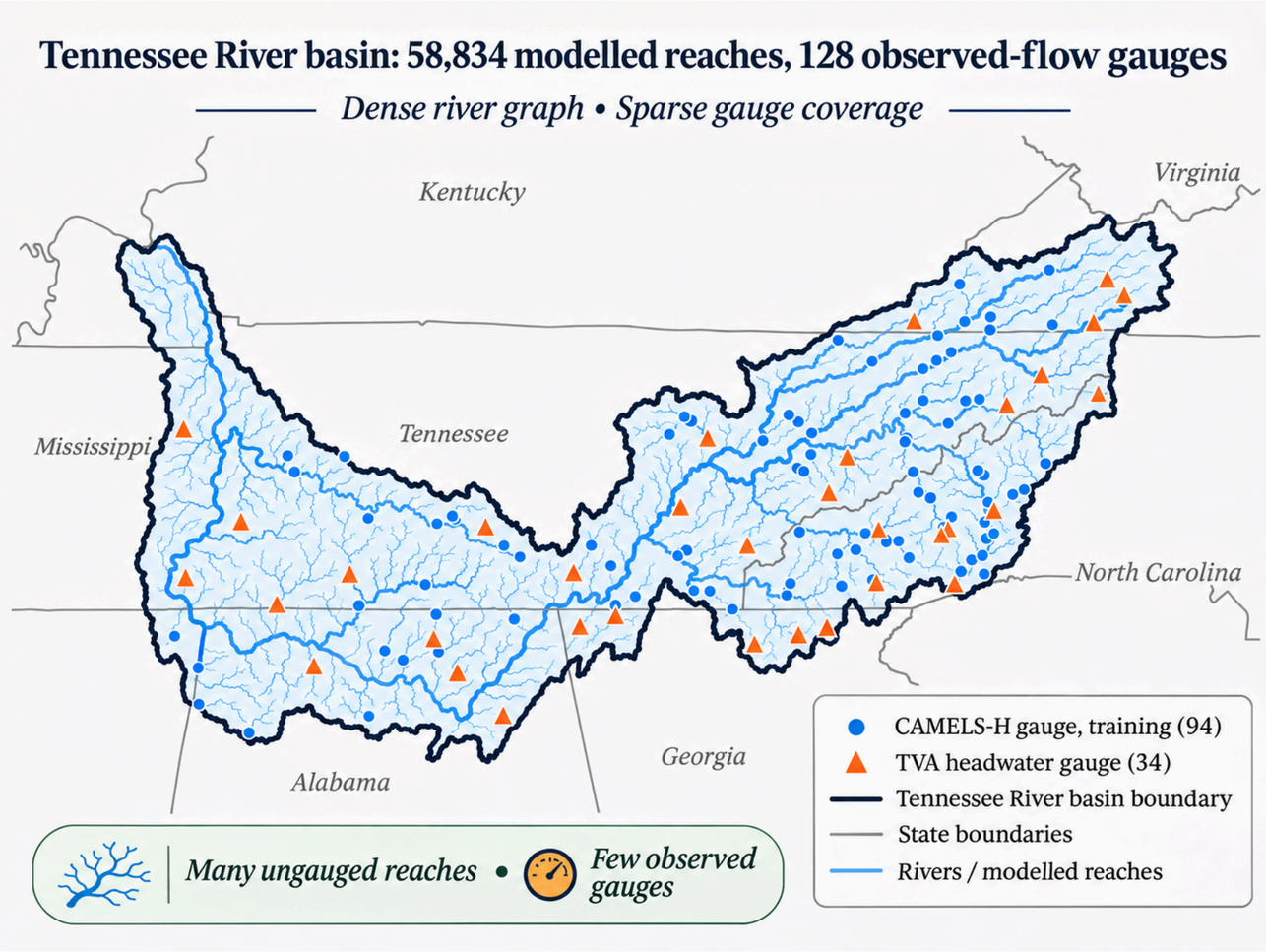
128

reaches with a public flow gauge

> 99.8%

of reaches are ungauged

The Tennessee River Basin

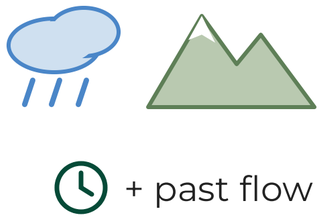


Forecast task

- **10-day** forecasts, **6-hourly** (40 steps)
- Every reach, one forward pass
- Future input: **issue-time GEFSv12 rain only**

Study 4: Model: reach, neighbours, routing

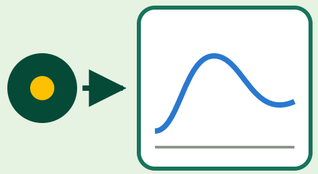
Inputs



+ past flow

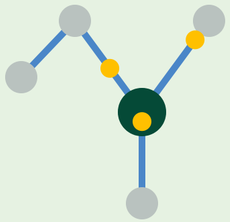
weather · rain forecast
catchment

1 · Remember



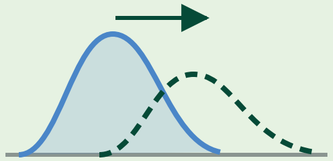
each reach reads
its own history

2 · Share



neighbours
exchange information

3 · Route



water travels
downstream

Flow at every reach

58,834

reaches

6-hourly, 10 days ahead

Where a gauge is visible:
recent observed flow sharpens the forecast, with upper quantiles for floods.

Two-stage training

A **Learn from simulation**

$$\mathcal{L}_A = \mathcal{L}\left(\hat{Q},\ Q^{\mathrm{NWM}}\right) \text{ at every reach}$$

B **Correct with observations**

$$\mathcal{L}_B = \underbrace{\mathcal{L}_{\mathrm{NSE}}}_{\text{gauges}} + \underbrace{\mathcal{L}_{\mathrm{pinball}}}_{q_{50},\, q_{90},\, q_{99}} + 0.3 \quad \underbrace{\mathcal{L}_{\mathrm{prior}}}_{\text{stay near NWM}}$$

Study 4: Forecast protocol and test design

One forecast rule for every model

Issue time · 00 UTC daily

Observed: AORC weather + USGS flow (36 days)	**Only the GEFSv12 control-member rain forecast**
history window	40 six-hour steps → 10 days

Periods

Train	GEFSv12 reforecast, 2000–2018
Model selection	2019
Test	**23 Sep 2020 – 22 Dec 2025: 1,917 daily issues**

Two tests at observed gauges

Gauged	Recent flow shown
Held-out	5 folds, recent flow never seen in training

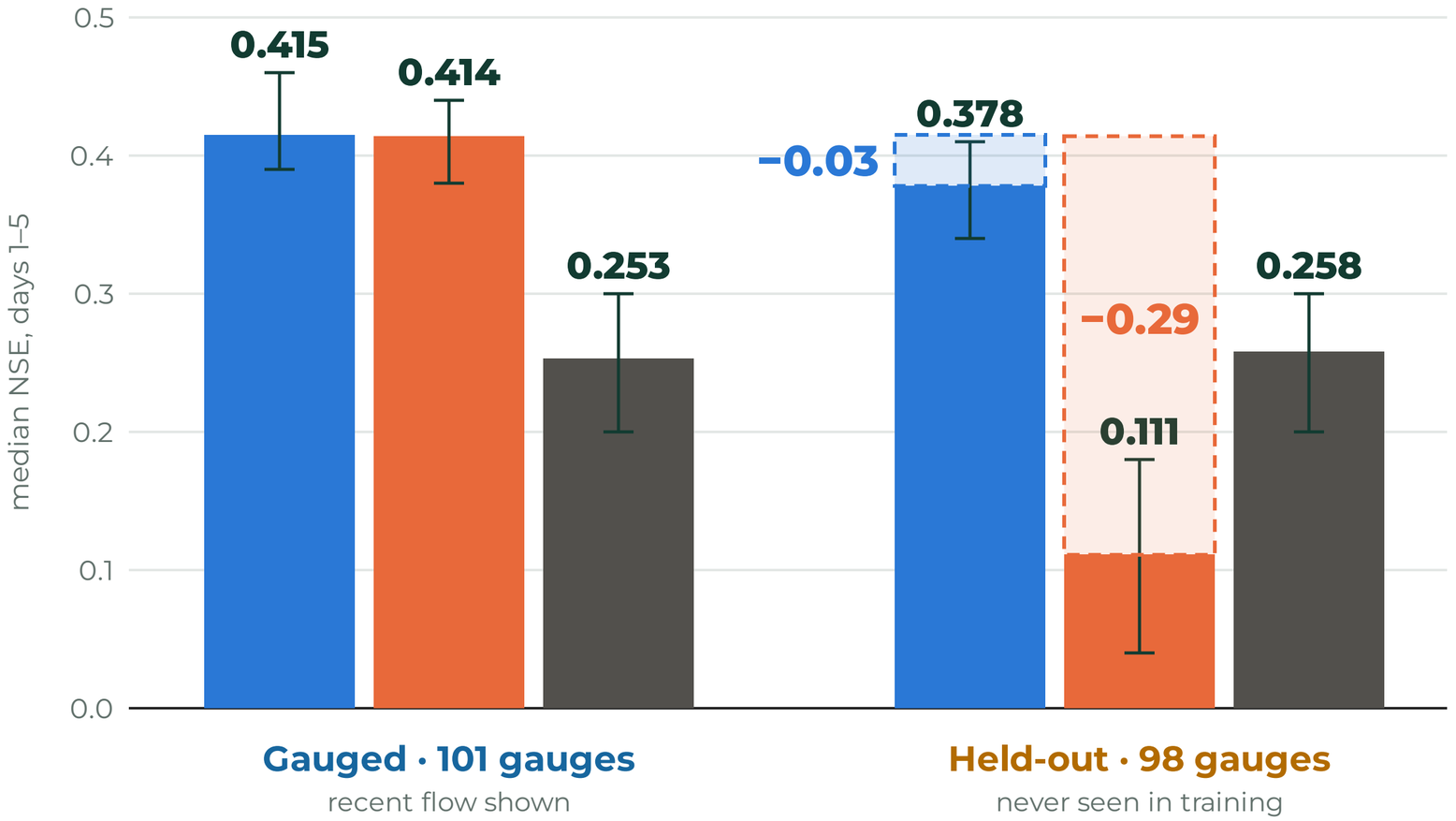
Same issues, gauges, masks, and GEFS rain for GNN and LSTM.

Baselines

- **Tuned basin LSTM** (256 units), same rain and masks
- **NWM operational forecast**: a reference; own forcing and assimilation

Score: per-gauge mean NSE over 24, 48, 72, 96, 120 h ("days 1–5"), median over gauges, 95% gauge-bootstrap interval.

Study 4: Key result, skill at gauges the model never saw



GNN **LSTM** (tuned)

NWM operational

Gauged

GNN and LSTM **tie**: 0.415 vs 0.414

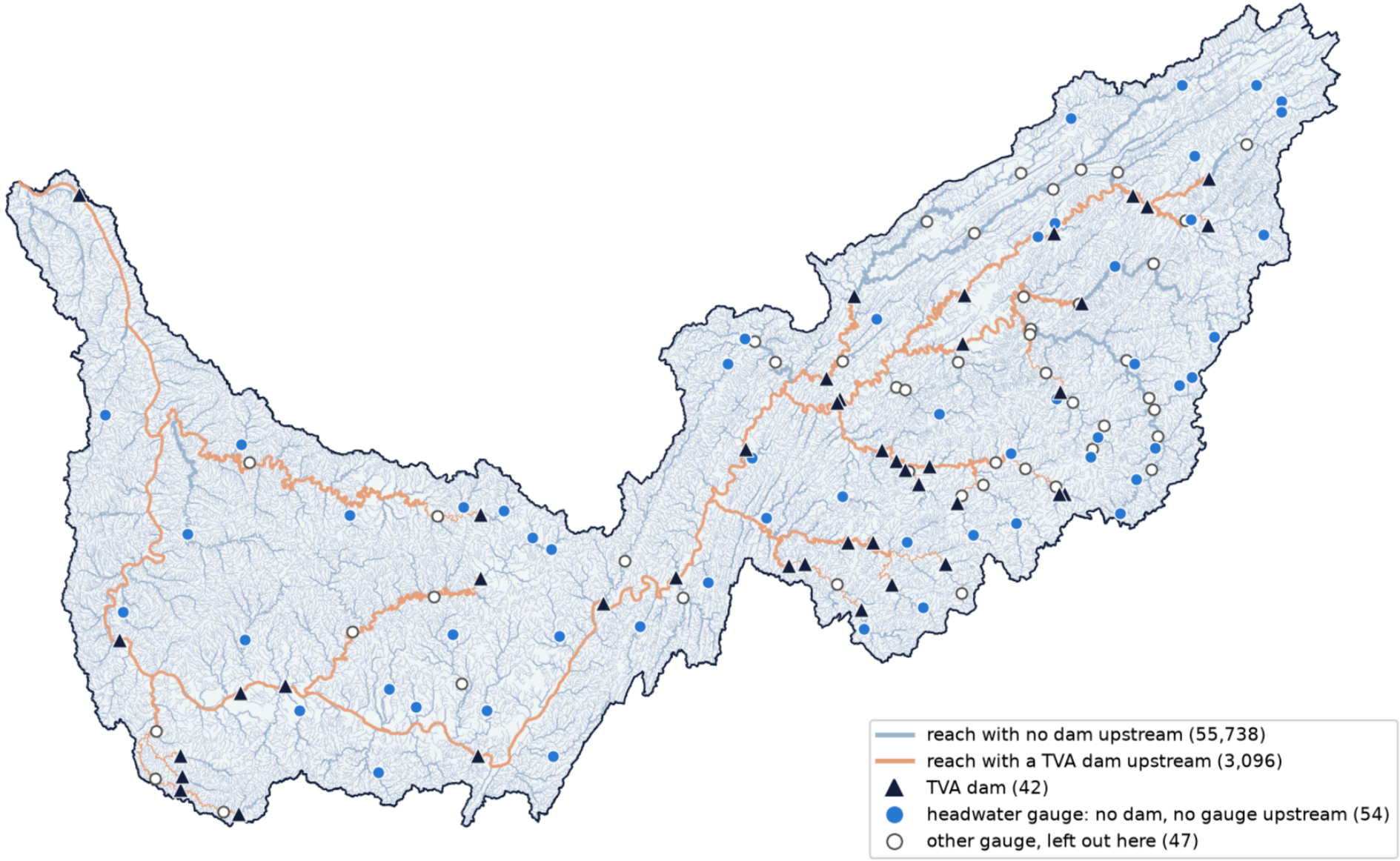
Held-out

GNN **0.378** (0.34–0.41)
LSTM 0.111 (0.04–0.18)
NWM 0.258

Whiskers: 95% gauge-bootstrap intervals. NWM assimilates USGS observations and may use held-out gauges, so it is a reference, not a controlled comparison.

Losing the gauge costs the GNN 0.03 NSE, the LSTM 0.29.

Study 4: A regulated basin: the headwater gauges



42

TVA dams regulate the main stems

54

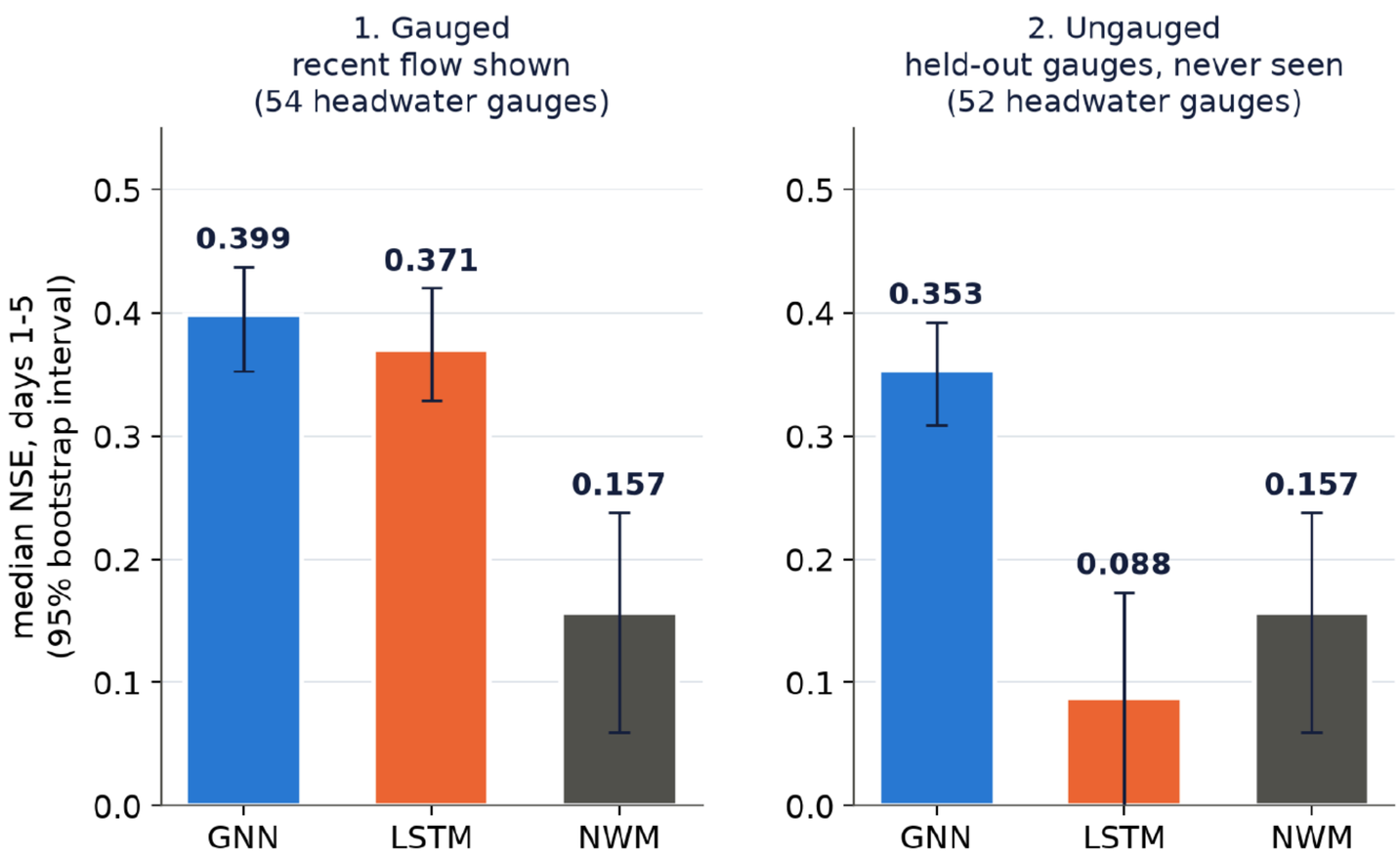
headwater gauges: no dam, no gauge upstream

95%

of reaches have no dam upstream

Study 4: Headwater basins without dams

Median NSE, days 1–5



Gauged

94%

of headwater gauges where the GNN beats NWM

Held-out

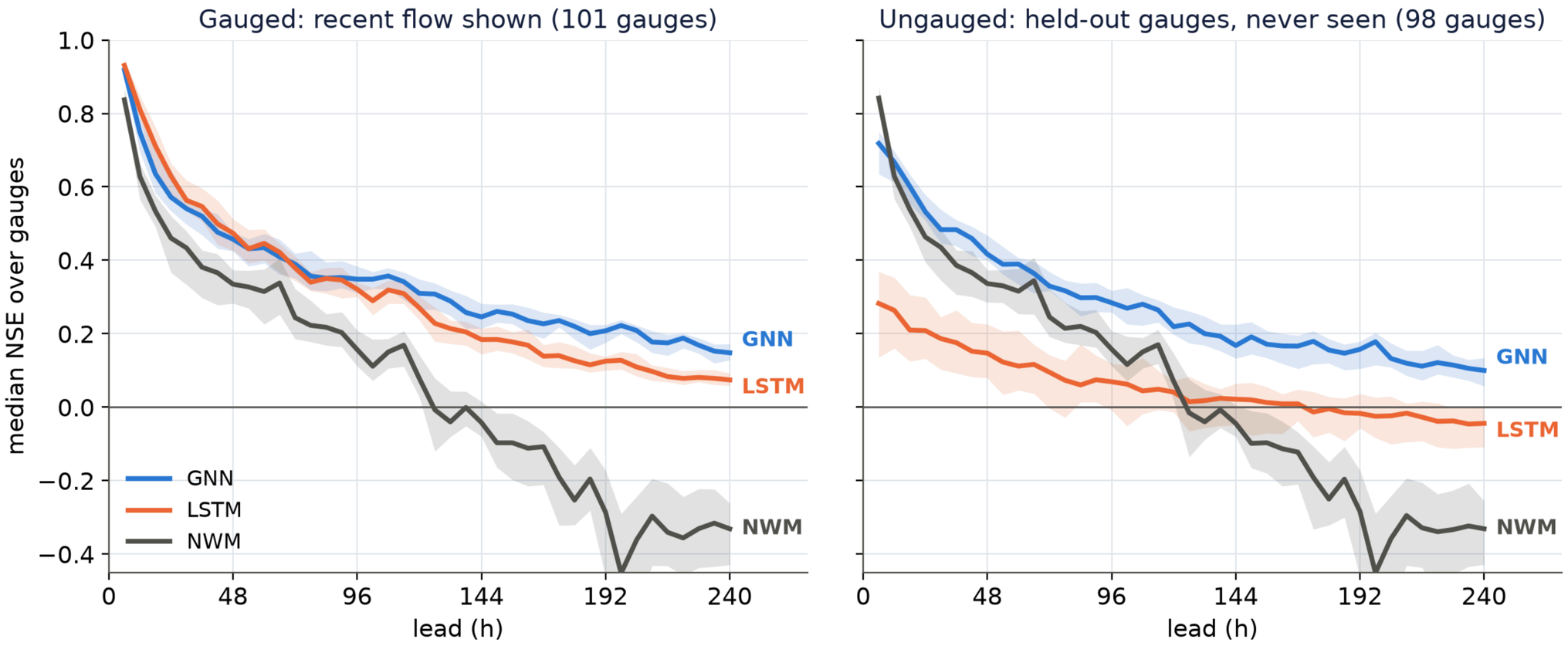
90%

of unseen gauges where the GNN beats the LSTM (87% vs NWM)

Without dams, the GNN is **ahead at gauges** and far ahead at unseen gauges.

Study 4: Lead time

Median NSE by lead time, 6 h → 10 days



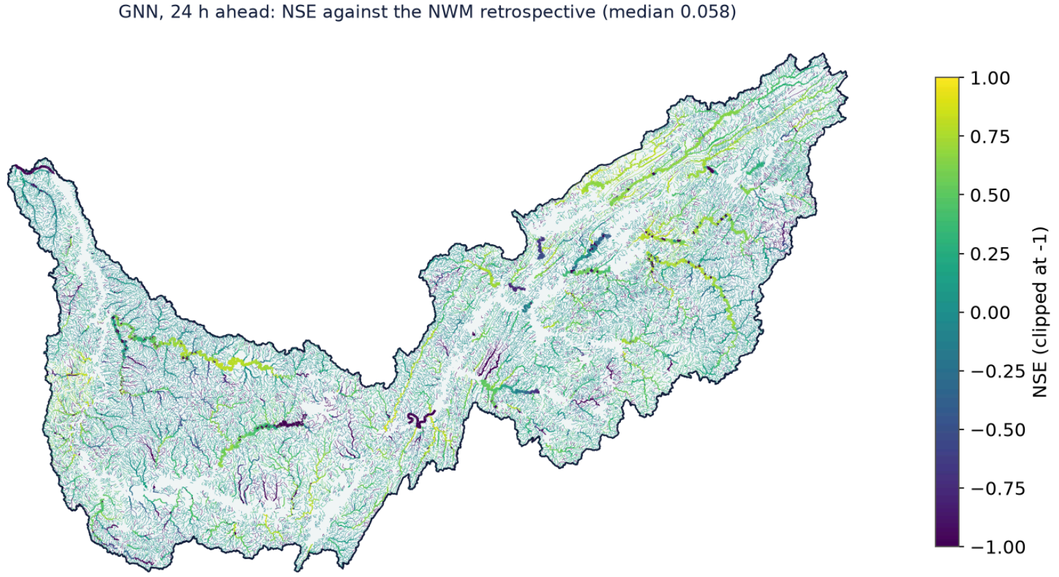
Gauged

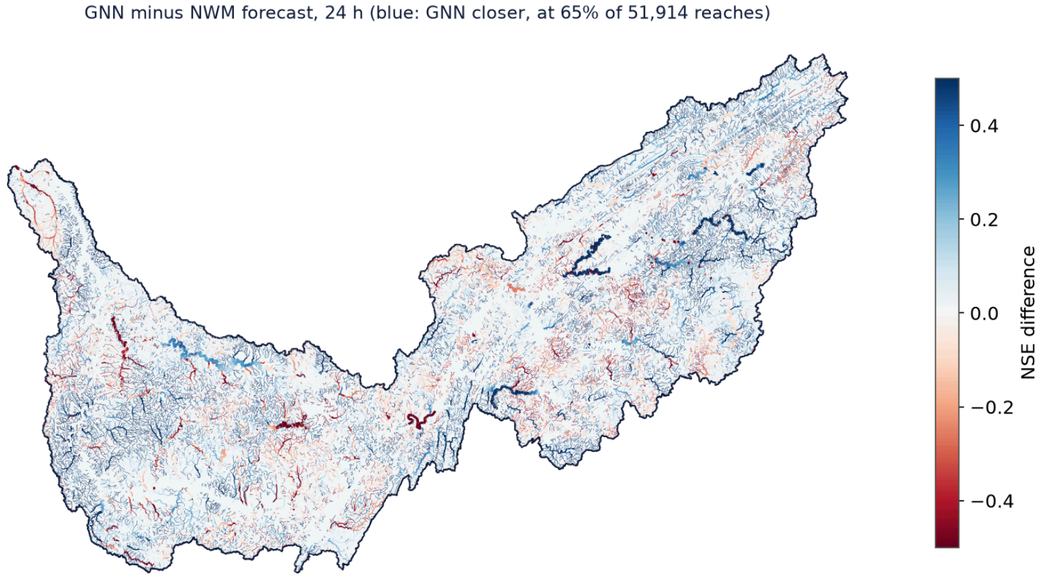
LSTM leads the first day. The **GNN leads from ~3 days**.

Held-out

The **GNN beats the LSTM at every lead**. NWM wins only at 6 h.

Study 4: Every reach, 24 h ahead





How often is the GNN closer?

vs NWM · 24 h

65%

of reaches: the GNN is closer

vs NWM · 72 h

70%

of reaches: the GNN is closer

vs LSTM · 24 h

88%

of reaches: the GNN is closer

The reference is a simulation, not observations.

Study 4: Conclusion

A river-network representation transfers to withheld gauges under a realistic protocol.

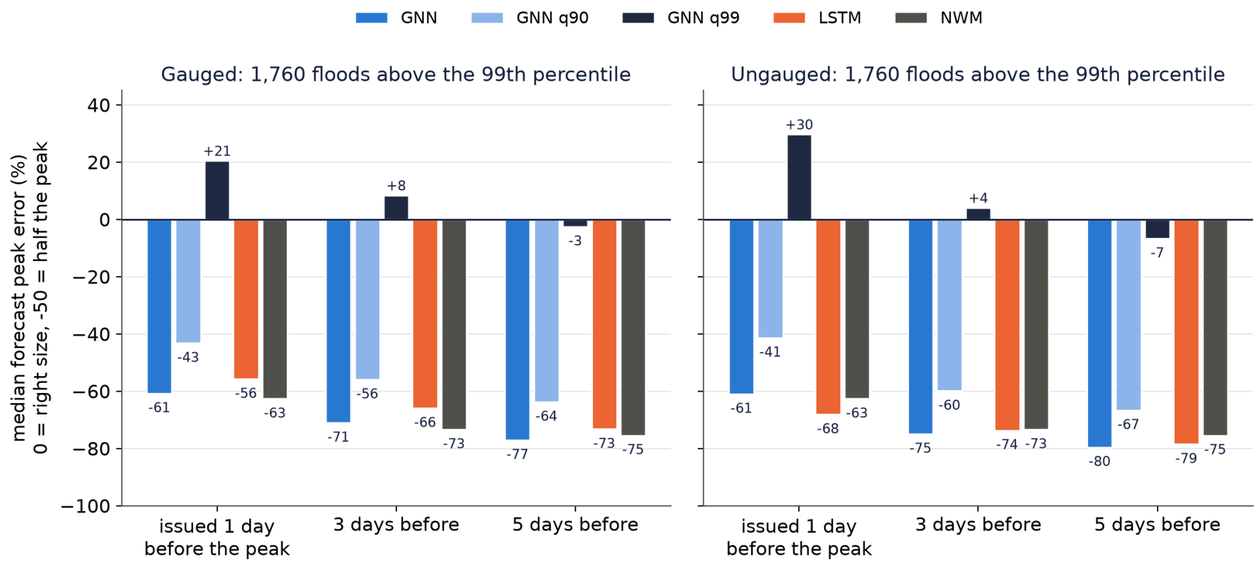
Where the GNN helps

- **Ungauged reaches:** keeps skill where no gauge was seen.
- **Longer leads:** strongest beyond ~3 days.
- **Natural-flow headwaters:** ahead even at gauges.

Limitations

- Floods: peaks too low; extremes missed.
- Scale: one basin; other regions untested.
- Attribution: graph, routing, or prior?

Limitation: flood peaks are too low



Peaks above the 99th percentile: **56–80% too low** for every model. Hurricane Helene (28–45×) was missed by all.

Next: tokenize the basin and pretrain representations at scale

What's next at ORNL

PROJECT

HydroORBIT

From one basin to a **pretrained, national** forecasting model

with Dan Lu, Xiao Wang, Vinh Tran

1

Tokenize the basin

Fields, river network and gauges become one shared input.

2

Pretrain at scale

Learn once from simulation and sparse observations, on Frontier.

3

Deliver forecasts that transfer

Inflow and flood forecasts at ungauged reaches, with uncertainty.

First, close the open questions: why the network transfers, flood peaks, transfer beyond Tennessee.

Summary

Hydrologic generalization depends on **how we represent the basin**.

Established

- **Spatial fields** beat lumped inputs, most in unseen basins.
- **BASINGRID** standardizes distributed data at national scale.
- **Learned basin embeddings** are reusable across predictors and transfer to unseen basins.
- **A river network** transfers to withheld gauges.

To be established

- **Flood peaks** along the river network.
- **Transfer beyond one region**, at national scale.
- **Pretrained, reusable** representations (HydroORBIT).

Next: tokenize the basin and pretrain representations at scale.

Thank you

Questions?

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